

The sensitivity of most condenser chamber-electrometer units is defined in this manner.

Example 6-12

The capacitance of a 100-R condenser chamber is 50 picofarads. The capacitance of the charger-reader used with the chamber is 10 picofarads. The volume of the chamber is 0.46 cm^3 . What is the sensitivity of the chamber? What reduction in voltage occurs when the chamber is exposed to 1 C/kg ?

$$\begin{aligned} \text{Sensitivity} &= \frac{V}{X} = \frac{1.29v}{C + C_e} \\ &= \frac{1.29(0.46 \text{ cm}^3)(10^{-6} \text{ m}^3/\text{cm}^3)}{(50 + 10) \times 10^{-12} \text{ farads}} \\ &= 989 \frac{\text{volts}}{\text{C/kg}} \\ \text{Sensitivity} &\simeq 1000 \frac{\text{volts}}{\text{C/kg}} \end{aligned}$$

Many types of exposure-measuring devices are available in which the chamber remains connected to an electrometer, a device for measuring small electric currents or charge by a shielded cable during exposure. While in the x-ray beam, the chamber may be made sensitive to radiation for a selected interval of time, and the exposure rate or cumulative exposure may be determined. A typical device of this type is illustrated in Margin Figure 6-11.

Radiation Quality

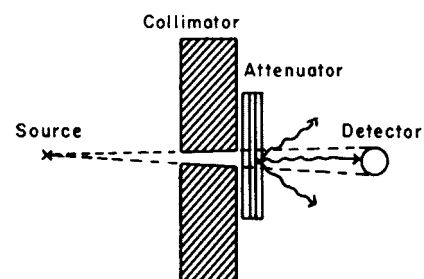
An x-ray beam is not described completely by stating the exposure or dose it delivers to a small region within an irradiated medium. The penetrating ability of the radiation, often described as the *quality* of the radiation, must also be known before estimates may be made of the following:

1. Exposure or dose rate at other locations within the medium
2. Differences in energy absorption between regions of different composition
3. Biological effectiveness of the radiation

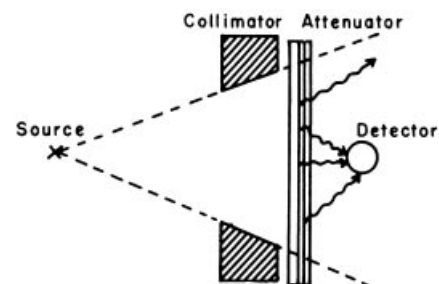
The spectral distribution of an x-ray beam depicts the relative number of photons of different energies in the beam. The quality of an x-ray beam is described explicitly by the spectral distribution. However, spectral distributions are difficult to measure or compute and are used only rarely to describe radiation quality. Usually, the quality of an x-ray beam is described by stating the half-value layer (HVL) of the beam, sometimes together with the potential difference (kilovolts peak) across the x-ray tube. Although other methods have been developed to describe the quality of an x-ray beam, the HVL is adequate for most clinical applications of x radiation.

■ HALF-VALUE LAYER

The HVL of an x-ray beam is the thickness of a material that reduces the exposure rate of the beam to half. Although the HVL alone furnishes a description of radiation quality that is adequate for most clinical situations, the value of a second parameter



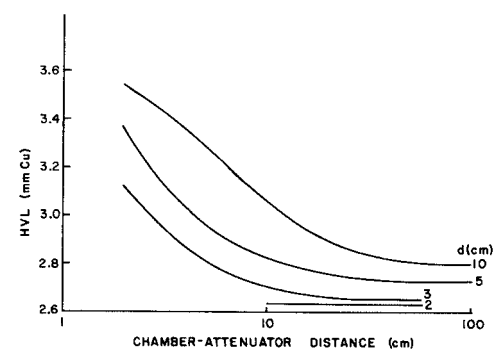
NARROW-BEAM GEOMETRY



BROAD-BEAM GEOMETRY

MARGIN FIGURE 6-12

Narrow-beam geometry is achieved by using a narrow beam and a large distance between the attenuator and the chamber used to measure radiation exposure. With narrow-beam geometry, few photons are scattered into the chamber by the attenuator. A significant number of photons are scattered into the chamber when attenuation measurements are made with broad-beam geometry.

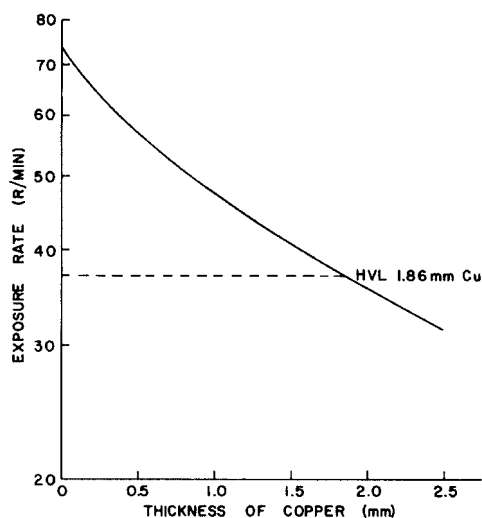


MARGIN FIGURE 6-13

Variation of the HVL with the diameter d of an x-ray beam at the attenuator for various distances between the attenuator and the chamber. The x-ray beam was generated at 250 kVp and filtered by 2 mm Cu. The attenuator was placed 20 cm below the x-ray tube. The geometry improves with decreasing diameter of the x-ray beam and increasing distance between attenuator and chamber. (From International Commission on Radiological Units and Measurements. *Physical Aspects of Irradiation*. National Bureau of Standards Handbook 85, 1964. Used with permission.)

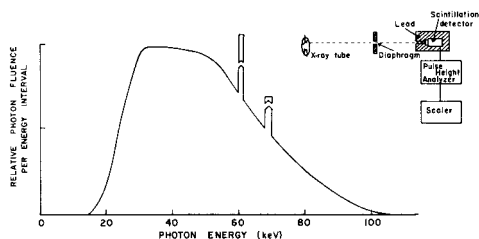
Half-value layer (HVL) is sometimes referred to as half-value thickness (HVT).

The homogeneity coefficient of an x-ray beam is the quotient of the thickness of the attenuator required to reduce the exposure rate to half divided by the thickness of attenuator required to reduce the exposure rate from half to one-fourth. The homogeneity coefficient is the ratio of the first and second HVLs.



MARGIN FIGURE 6-14

Exposure rate for an x-ray beam as a function of the thickness of copper added to the beam. The HVL of the beam is 1.86 mm Cu.



MARGIN FIGURE 6-15

Energy distribution of x-ray photons in a beam generated at 105 kVp and filtered by 2 mm Al. Equipment used to measure the spectra of primary x rays is diagramed in the inset. (From Epp, E., and Weiss, H. *Phys. Med. Biol.* 1966; 11:225. Used with permission.)

(e.g., the peak potential difference across the x-ray tube or the homogeneity coefficient of the x-ray beam) is sometimes stated with the HVL. Once the HVL of an x-ray beam is known together with the exposure or dose rate at a particular location, the absorbed dose rate may be computed at other locations within an irradiated medium.

HVLs are measured with solid “absorbers” (more correctly, attenuators) such as thin sheets of aluminum, copper, or lead of uniform thickness. HVLs may be measured by placing attenuators of different thickness but constant composition between the x-ray tube and an ionization chamber designed to measure radiation exposure. The distribution of photon energies changes as attenuators are added to an x-ray beam, and the response of the chamber should be independent of these changes. Measurements of HVL always should be made with narrow-beam geometry. Narrow-beam geometry (sometimes referred to as “good” geometry) ensures that the only photons that enter the chamber are primary photons that have been transmitted by the attenuator. Requirements for narrow-beam geometry are satisfied if the chamber is positioned far from the attenuator, and an x-ray beam with a small cross-sectional area is measured. The cross-sectional area of the beam should be just large enough to deliver a uniform exposure over the entire sensitive volume of the chamber. Conditions of narrow-beam and broad-beam “poor” geometry are depicted in Margin Figure 6-12. Measurements of HVL under these conditions are compared in Margin Figure 6-13. With broad-beam geometry, more photons are scattered into the detector as the thickness of the attenuator is increased. Consequently, HVLs measured with broad-beam geometry are greater than those measured under conditions of narrow-beam geometry. HVLs measured under conditions of broad-beam geometry are used for the calculation of shielding requirements for walls surrounding x-ray equipment.

Narrow-beam geometry may be achieved by placing the attenuator midway between the x-ray target and the chamber, with the chamber at least 0.5 m from the attenuator. The area of the field should be no larger than a few square centimeters. Objects that might scatter photons into the chamber should be removed from the vicinity of the x-ray beam. Even greater accuracy can be attained by measuring the HVL for fields of different cross-sectional area and extrapolating the measured HVLs to a field size of zero.³¹ However, this procedure is not necessary for routine clinical dosimetry.

An attenuation curve for x rays in copper is shown in Margin Figure 6-14. The HVL is 1.86 mm Cu for the x-ray beam described in the figure. A complete attenuation curve is not required for routine dosimetry. Instead, thicknesses of the attenuator are identified that reduce the exposure rate to slightly more than half (50% to 55%) and slightly less than half (45% to 50%). These data are plotted on a semilogarithmic graph and connected by a straight line. The approximate HVL is given by the intersection of this straight line with a horizontal line drawn through the ordinate (y axis) at half the exposure rate measured with no attenuator added to the beam.

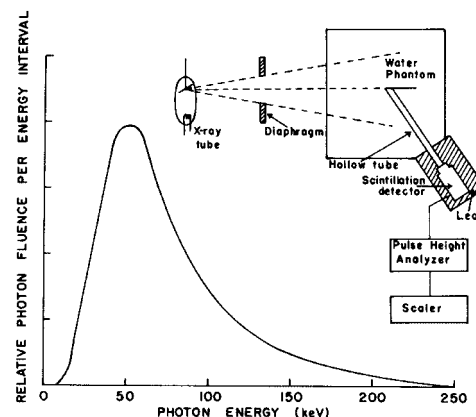
VARIATION IN QUALITY ACROSS AN X-RAY BEAM

The exposure rate on the side of an x-ray beam nearest the anode is less than the exposure rate on the cathode side because x-rays on the anode side are attenuated by a greater thickness of target material before they emerge from the anode. This variation in exposure rate across an x-ray beam is termed the *heel effect* (see Chapter 4). The greater filtration on the anode side increases the HVL of the x-ray beam from the cathode to the anode side. The decrease in exposure rate from cathode to anode across an x-ray beam is less noticeable at depths within a patient because the radiation on the anode side is more penetrating.

SPECTRAL DISTRIBUTION OF AN X-RAY BEAM

The spectral distribution for a beam of x or γ rays may be computed from an attenuation curve measured under conditions of narrow-beam geometry. A variety of curve-fitting techniques may be applied to the attenuation curve to obtain equations that are used to compute the spectral distribution. The accuracy of this method of obtaining a spectral distribution is limited by the accuracy of the curve-fitting technique and the accuracy with which the attenuation curve is measured.

Most measurements of x- and γ -ray spectra are made with a scintillation or semiconductor detector. The height of a voltage pulse from one of these detectors varies with the energy deposited in the detector by an x or γ ray. The pulses are sorted by height in a pulse height analyzer and counted in a scaler. The recorded counts are plotted as a function of pulse height to furnish a *pulse-height distribution*, which reflects the energy distribution of photons impinging upon the detector. To portray the energy distribution accurately, the pulse-height distribution must be corrected for statistical fluctuations in the energy distribution and for incomplete absorption of photons in the detector. Measured spectra for primary and scattered x rays are shown in Margin Figures 6-15 and 6-16.



MARGIN FIGURE 6-16

Energy distribution of scattered radiation at 15-cm depth in a water phantom exposed to a 200-cm² x-ray beam generated at 250 kVp. Equipment used to measure the spectra of scattered x rays is diagramed in the inset. (From Hettinger, G., and Liden, K. *Acta Radiol.* 1960; 53:73. Used with permission.)

PROBLEMS

- *6-1. The photon flux density is 10^9 photons/(m²-sec) for a beam of γ rays. One-fourth of the photons have an energy of 500 keV, and three-fourths have an energy of 1.25 MeV. What is the energy flux density of the beam? If the photon flux density is constant, what is the energy fluence over an interval of 10 seconds?
- *6-2. During an exposure of 0.02 coulomb/kg, how much energy is absorbed per kilogram of air?
- *6-3. The energy absorption coefficient of air is 2.8×10^{-3} m²/kg for photons of 1.0 MeV. What is the energy fluence required for an exposure of 0.03 coulomb/kg? What is the photon fluence for this exposure?
- *6-4. How many ion pairs are produced in 1 cm³ of air during an exposure of 0.03 coulomb/kg?
- *6-5. Water is 89% oxygen (gram-atomic mass, 15.999) and 11% hydrogen (gram-atomic mass, 1.008) by weight. Compute the effective atomic number of water.
- *6-6. A thimble chamber with an air-equivalent wall receives an exposure of 0.015 coulomb/kg in 1 minute. The volume of the chamber is 0.46 cm³. What is the ionization current from the chamber?
- *6-7. A condenser ionization chamber has a sensitivity of 7750 V/(C-kg⁻¹). The volume of the chamber is 0.46 cm³. The capacitance of the chamber is six times the capacitance of the charger-reader. What is the capacitance of the chamber?
- *6-8. A miniature ionization chamber has a sensitivity of 3876 V/(C-kg⁻¹). The chamber is discharged by 100 V during an exposure to x radiation. What exposure did the chamber receive?
- 6-9. A Victoreen condenser chamber receives identical exposures in New Orleans (sea level) and in Denver (5000 ft above sea level). Is the deflection of the platinum wire lower or higher in Denver? Why?
- 6-10. What is meant by the energy dependence of an air-wall thimble chamber?
- *6-11. An organ with a mass of 10 g receives a uniform dose of 10 cGy. How much energy is absorbed in each mass of tissue in the tumor, and how much energy is absorbed in the entire mass?
- *6-12. A particular type of lesion recedes satisfactorily after receiving 55 Gy from orthovoltage x rays. When the lesion is treated with ⁶⁰Co γ radiation, a dose of 65 Gy is required. Relative to orthovoltage x rays, what is the RBE of ⁶⁰Co radiation for treating this type of lesion?
- *6-13. An individual who ingests a radioactive sample receives an estimated dose of 15 mGy to the testes: 11 mGy is delivered by β -particles, and 4 mGy is delivered by γ -radiation. What is the dose to the testes in millisievert?

*For problems marked with an asterisk, answers are provided on p. 491.

- *6-14. The specific heat of graphite is 170 cal/kg°C. A uniform dose of 10 Gy is delivered to a graphite block insulated from its environment. What is the rise in temperature of the block?
- *6-15. With the assumption that $G = 15.4$, how many Fe^{2+} ions are oxidized if an absorbed dose of 150 Gy is delivered to a 10-mL solution of ferrous sulfate? Assume that the density of ferrous sulfate is 10^{-3} kg/mL.
- 6-16. Explain clearly what is meant by the energy dependence of a radiation dosimeter.
- *6-17. Attenuation measurements for an x-ray beam from a 120-kVp x-ray generator yield the following results:

Added Filtration (mm Al)	Percent Transmission
1.0	60.2
2.0	41.4
3.0	30.0
4.0	22.4
5.0	16.9

Plot the data on semilogarithmic graph paper and determine the following:

- The first HVL
- The second HVL
- The homogeneity coefficient of the x-ray beam.

SUMMARY

- Intensity is energy per unit area per unit time.
- Fluence is particles, photons or energy per unit area.
- Flux is time rate of change of fluence.
- Exposure is a measure of the charge liberated per unit mass of air by ionizing radiation.
- Dose is energy absorbed in any material per unit mass.
- Calculation of RBE dose includes a conversion factor that is based upon experiment.
- Dose equivalent includes a quality factor that accounts for differences in biological harm caused by equal doses of different types of radiation. It assumes that the dose to a point in the organ is known.
- Equivalent dose is similar to dose equivalent, but applies slightly different weighting factors and uses the average dose to an organ or body part.
- Effective dose equivalent used weighting factors to account for biological harm to the individual and genetic effects in the next two generations caused by irradiation of different body regions. It has been superseded by the quantity known as equivalent dose.
- Effective dose converts average organ and body region doses to a measure of overall detriment to the individual and to future generations through genetic effects. It also takes into account the type of radiation received. It is based upon radiation weighting factors W_r and tissue weighting factors W_t that were derived in the early 1990s from analysis of lifetime follow-up of atom bomb survivors.
- Quantities and their units are given below:

Quantity	SI Unit	Traditional Unit
Exposure	coulomb/kg	roentgen
Dose	gray (Gy)	rad
RBE dose	sievert (SV)	rem
Dose equivalent	sievert (SV)	rem
Equivalent dose	sievert (SV)	rem
Effective dose equivalent	sievert (SV)	rem
Effective dose	sievert (SV)	rem

- Methods of measurement of radiation dose include:
 - Calorimetry
 - Photographic methods
 - Chemical
 - Scintillation
 - Thermoluminescence
 - Solid-state detectors
 - Ionization chambers

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INTERACTION OF X AND γ RAYS IN THE BODY

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OBJECTIVES

By studying this chapter, the reader should be able to:

- Explain the origin of the f factor, and determine the dose (Gy) to a medium from knowledge of the exposure (C/kg).
- Describe the attenuation of x rays in different tissues as a function of x-ray energy.
- Discuss the properties of various tissues that permit their distinction in x-ray images.
- Define changes in radiation dose at the interface between bone and soft tissue.
- Identify the applications of high-voltage and low-voltage radiography.
- Characterize the properties and applications of contrast media.

INTRODUCTION

The dominant mode of interaction of x and γ rays in a region of the body varies with the energy of the photons and the effective atomic number and electron density (electrons/kilogram) of the region. For the purpose of discussing interactions, the body may be divided into regions of (1) fat, (2) muscle (or soft tissue excluding fat), (3) bone, and (4) air-filled cavities.

F FACTOR

An exposure X of 1 coulomb/kilogram (C/kg) provides an absorbed dose D_{air} in air of 33.85 gray (Gy) (see Chapter 6):

$$D_{\text{air}}(\text{Gy}) = 33.85X \text{ (C/kg)}$$

The dose to a medium such as soft tissue is related to the dose in air at the same location multiplied by the ratio of energy absorption in the medium to that in air:

$$D_{\text{med}} = D_{\text{air}} \frac{[(\mu_{\text{en}})_m]_{\text{med}}}{[(\mu_{\text{en}})_m]_{\text{air}}} \quad (7-1)$$

$$D_{\text{med}}(\text{Gy}) = 33.85X \frac{[(\mu_{\text{en}})_m]_{\text{med}}}{[(\mu_{\text{en}})_m]_{\text{air}}} \quad (7-2)$$

with X in units of C/kg.

In Eq. (7-1), $[(\mu_{\text{en}})_m]_{\text{med}}$ is the mass energy absorption coefficient of the medium for photons of the energy of interest. This coefficient describes the rate of energy absorption in the medium. The expression $[(\mu_{\text{en}})_m]_{\text{air}}$ describes the rate of energy absorption in air for photons of the same energy. In Eq. (7-2), the expression may be simplified to

$$D_{\text{med}}(\text{Gy}) = (f)(X) \quad (7-3)$$

where

$$f = 33.85 \frac{[(\mu_{\text{en}})_m]_{\text{med}}}{[(\mu_{\text{en}})_m]_{\text{air}}} \quad (7-4)$$

The expression denoted as f is known as the f factor. This factor varies with the nature of the absorbing medium and the energy of the radiation.

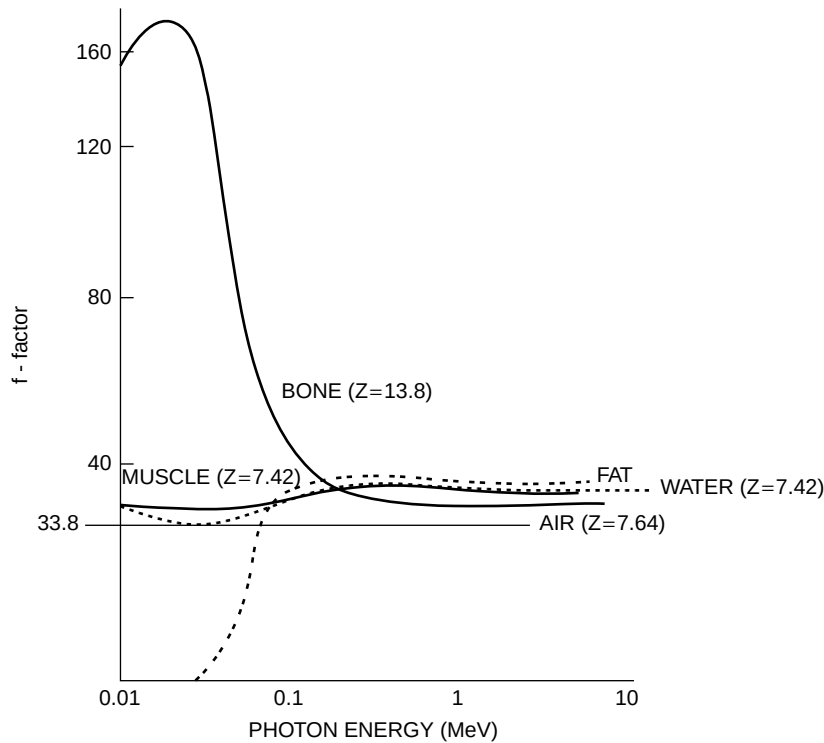
The f factor is used to compute the absorbed dose D in gray in a medium receiving an exposure X in coulombs per kilogram. The f factor is plotted in Figure 7-1 for air, fat, muscle, and compact bone as a function of photon energy.

D_{air} was originally defined as $D_{\text{air}} = 0.869X$, where D_{air} was expressed in rads and the exposure X was expressed in roentgens. One roentgen was defined as “the amount of x or γ rays that produces 2.58×10^{-4} coulombs of charge of either sign in 1 kg of dry air.” The roentgen, the earliest quantitative measure of radiation, is no longer sanctioned by international scientific commissions as an official unit of exposure.



MARGIN FIGURE 7-1

(Top) Photograph of Wilhelm Conrad Röntgen as a student. (Bottom) Röntgen in 1906 while director of the Institute of Physics at the University of Munich.

**FIGURE 7-1**

The f factor for conversion between exposure (C/kg) and absorbed dose (Gy) for air, water, and different constituents of the body, plotted as a function of photon energy. Curves do not extend beyond 3 MeV because radiation exposure (and therefore the f factor) is not applicable to photons of higher energy. Because of the varying composition of fat, the dotted curve for this tissue constituent is only approximately correct.

This plot illustrates why image contrast is reduced at higher photon energies where interactions are primarily Compton scattering rather than photoelectric absorption.

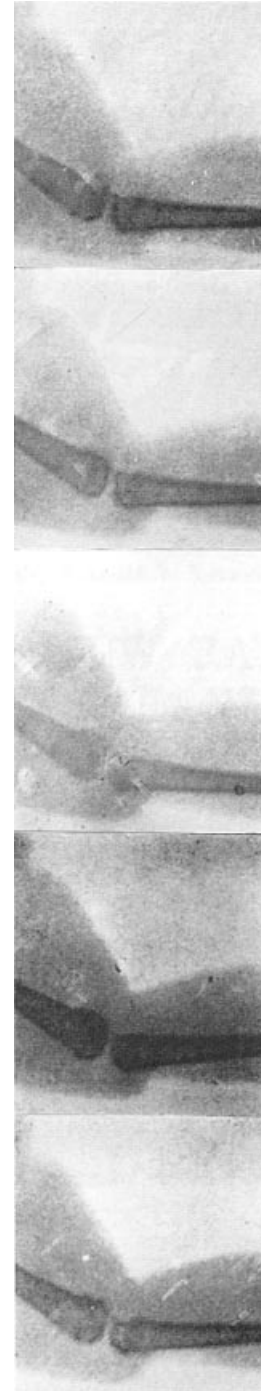
For all photon energies for which exposure (and therefore the f factor) is defined, an exposure of 1 C/kg provides an absorbed dose of 33.85 Gy in air. Consequently, the f factor equals 33.85 for air and is independent of photon energy.

■ ATTENUATION OF X AND γ RAYS IN TISSUE

A simplified model of the human body consists of three different body tissues: fat, muscle, and bone. Air is also present in the lungs, sinuses and gastrointestinal tract, and a contrast agent may be used to accentuate the attenuation of x rays in a particular region. The elemental composition of the three body tissues, together with their percent mass composition, are shown in Table 7-1. Selected physical properties of the tissues are included in Table 7-2. Mass attenuation coefficients for different tissues as a function of photon energy are shown in Figure 7-2.

In Table 7-2, the data for muscle are also approximately correct for other soft tissues such as collagen, internal organs (e.g., liver and kidney), ligaments, blood, and cerebrospinal fluid. These data are also close to the data for water, because soft tissues, including muscle, are approximately 75% water, and body fluids are 85% to 100% water. The similarity of soft tissues suggests that conventional x-ray imaging yields poor discrimination among them. Sometimes a contrast agent can be used to accentuate the small intrinsic differences in x-ray attenuation among soft tissues.

Compared with other tissue constituents, fat has a greater concentration of low- Z elements, especially hydrogen. Therefore, fat has a lower density and effective atomic number compared with muscle and other soft tissues. Below about 35 keV, x rays interact in fat and other soft tissues predominantly by photoelectric interactions. These

**MARGIN FIGURE 7-2**

First x-ray "movie" showing 5 views of a frog's leg filmed in 1897.¹



MARGIN FIGURE 7-3
Advertisement for x-ray studio.³

As described earlier, the common form of hydrogen (protium) contains no neutrons; hence atoms of this form of hydrogen have one electron for each nucleon (proton). For most other nucleons, including other forms (isotopes) of hydrogen, the ratio of electrons to nucleons (protons and neutrons) is 1:2 or less.

TABLE 7-1 Percent Mass Composition of Tissue Constituents⁴⁻⁵

% Composition (by Mass)	Adipose Tissue	Muscle (Striated)	Water	Bone (Femur)
Hydrogen	11.2	10.2	11.2	8.4
Carbon	57.3	12.3		27.6
Nitrogen	1.1	3.5		2.7
Oxygen	30.3	72.9	88.8	41.0
Sodium		0.08		
Magnesium		0.02		7.0
Phosphorus		0.2		7.0
Sulfur	0.06	0.5		0.2
Potassium		0.3		
Calcium		0.007		14.7

interactions vary with Z^3 of the tissue. This dependence on Z yields modest image contrast among tissues of slightly different composition (e.g., fat and muscle) when low-energy x rays are used. The contrast disappears with higher-energy x rays that interact primarily by Compton interactions, since these interactions do not vary with Z . Low-energy x rays are used to accentuate subtle differences in soft tissues (e.g., fat and other soft tissues) in applications such as breast imaging (mammography) where the object (the breast) provides little intrinsic contrast. When images are desired of structures with high intrinsic contrast (e.g., the chest where bone, soft tissue, and air are present), higher-energy x rays are used. These x rays suppress x-ray attenuation in bone which otherwise would create shadows in the image that could hide underlying soft-tissue pathology.

When compared with muscle and bone, fat has a higher concentration of hydrogen ($\sim 11\%$) and carbon ($\sim 57\%$) and a lower concentration of nitrogen ($\sim 1\%$), oxygen (30%), and high- Z trace elements ($<1\%$) (Table 7-1).^{4,5} Hence, the effective atomic number of fat ($Z_{\text{eff}} = 5.9$ to 6.3) is less than that for soft tissue ($Z_{\text{eff}} = 7.4$) or bone ($Z_{\text{eff}} = 11.6$ to 13.8). Because of its lower Z_{eff} , low-energy photons are attenuated less rapidly in fat than in an equal mass of soft tissue or bone. The reduced attenuation in fat yields a lower f factor for low-energy photons in this body constituent (Figure 7-1). The attenuation of x and γ rays in fat may be estimated from attenuation measurements in mineral oil or polyethylene because the effective atomic numbers, densities, and electron densities of these materials are close to those for fat.⁶⁻⁸

X and γ rays of higher energy interact primarily by Compton scattering with a probability that varies with the electron density of the attenuating medium but not with the atomic number. The electron density of hydrogen is about twice that of other elements. Because more hydrogen is present in fat than in other body constituents, more Compton interactions occur in fat than in an equal mass of muscle or bone. For photons of intermediate energy, therefore, the f factor for fat exceeds that for other body constituents (Figure 7-1).

Hydrogen is absent from air but contributes about 10% of the weight of muscle. Consequently, the electron density is greater for muscle than for air, and the f factor for muscle exceeds that for air (Figure 7-1).

TABLE 7-2 Properties of Tissue Constituents of the Human Body

Material	Effective Atomic Number	Density (kg/m^3)	Electron Density (electrons/kg)
Air	7.6	1.29	3.01×10^{26}
Water	7.4	1.00	3.34×10^{26}
Muscle	7.4	1.00	3.36×10^{26}
Fat	5.9–6.3	0.91	$3.34\text{--}3.48 \times 10^{26}$
Bone	11.6–13.8	1.65–1.85	$3.00\text{--}3.10 \times 10^{26}$

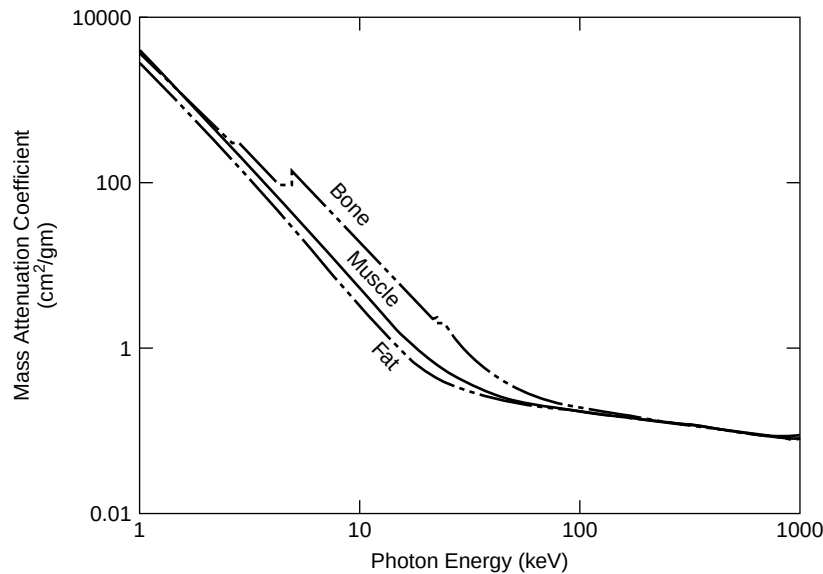


FIGURE 7-2
Mass attenuation coefficients of tissues.²

The effective atomic number and physical density are greater for bone than for soft tissue. Hence, x and γ rays are attenuated more rapidly in bone than in an equal volume (not necessarily mass) of soft tissue. This effect reduces the absorbed dose to structures beyond bone. The absorbed dose to soft tissue immediately adjacent to or enclosed within bone may be increased by photoelectrons liberated as photons interact with high-Z atoms (e.g., phosphorus and calcium) in bone.

Compared with muscle and fat, bone contains less hydrogen and therefore its electron density is slightly less as well. For this reason, the *energy absorbed per gram* of bone is slightly less than the energy absorbed per gram of muscle or fat exposed to photons of intermediate energy. As shown in Figure 7-1 the *f* factor for bone is less than the *f*-factor for either muscle or fat and intermediate photon energies. However, the physical density of compact bone is almost twice the density of fat or muscle. Therefore, the *energy absorbed per unit volume* of compact bone is almost twice that absorbed in an equal volume of fat or muscle exposed to x and γ rays of intermediate energy.

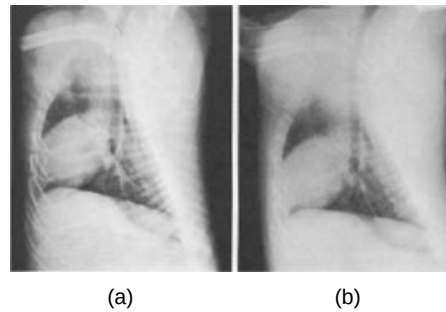
In a radiograph obtained by exposure of film to high-energy photons (e.g., 4 MV x rays), images of bone are displayed as regions of reduced optical density (i.e., they are more transparent to visible light). Hence the number of photons transmitted by bone is less than the number transmitted by an equal thickness of soft tissue. Although the energy absorbed per unit mass (absorbed dose) is less in bone than in soft tissue exposed to high-energy x rays, the transmission of photons through bone is less because the physical density of bone is greater than that for soft tissue.

■ DOSE TO SOFT TISSUE BEYOND BONE

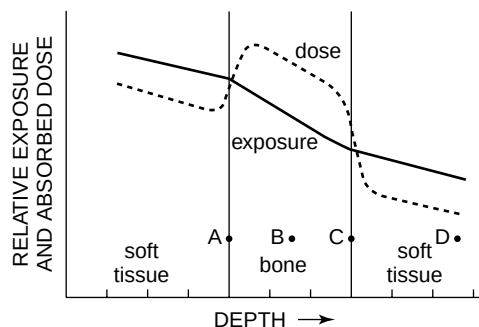
The dose delivered to soft tissue by x and γ rays is reduced by bone interposed between the soft tissue and the surface. The reduction in absorbed dose to soft tissue is influenced by:

1. Increased attenuation of primary photons in bone, caused by the higher atomic number and density of this tissue constituent.
2. Changes in the amount of radiation scattered to soft tissue beyond the bone, which depend on many factors, including field size, quality of radiation, and the distance between the bone and the soft tissue.

The effect of bone upon the radiation exposure and absorbed dose at various depths within a patient exposed to diagnostic x rays is illustrated in the margin.⁹ The radiation exposure is reduced at locations B, C, and D by bone interposed between the locations and the surface. The reduction primarily reflects the increased attenuation of photons in overlying bone. The dose absorbed in bone is increased at A, B, and C because the attenuation of diagnostic x rays increases with the atomic

**FIGURE 7-3**

Radiographs of the chest. **A:** 80 kVp, 1-mm Al filter. **B:** 140 kVp, 1-mm Cu filter.

**MARGIN FIGURE 7-4**

Radiation exposure and absorbed dose versus depth in soft tissue containing bone.

Chest radiography at a lower kVp may be desirable in cases where a broken rib is suspected.

X-ray computed tomography has alleviated the need for pneumoencephalography, a procedure that is so painful and frightening to patients that it is almost barbaric.

number of the attenuating medium. The absorbed dose is reduced to soft tissue beyond bone because more photons are removed from the beam by the overlying bone.

HIGH-VOLTAGE RADIOGRAPHY

Radiographic images obtained with x rays generated below 100 kVp exhibit high contrast between soft tissue and bone. In a radiograph of a chest exposed to 80-kVp x rays (Figure 7-3A), the image of bone obscures the visibility of the trachea. Shadows cast by bone may be reduced by increasing the voltage applied to the x-ray tube and adding filtration to the x-ray beam. The radiograph in Figure 7-3B was obtained with x rays generated at 140 kVp and filtered by 1 mm Cu. The trachea, lung, and retrocardiac markings are displayed more clearly in this radiograph than in Figure 7-3A.

High-voltage x-ray beams are used in radiology primarily for:

- study of air-filled structures such as the chest, larynx, and paranasal sinuses;
- myelography (introduction of air, gas, or other contrast agent into the subarachnoid space of the spinal column);
- pneumoencephalography (introduction of air or inert gas into the subarachnoid space of the spinal column and into the ventricular system of the brain);
- study of the gastrointestinal tract with a contrast agent when retention of some tissue differentiation is desired in structures containing the agent.

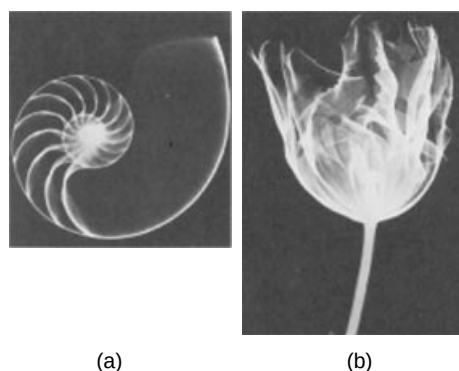
Disadvantages of high-voltage radiography include:

- a reduction in contrast between adjacent soft tissues;
- a reduction in radiographic detail caused by an increased amount of scattered radiation;
- reduction in the ability of grids to remove scattered radiation.

LOW-VOLTAGE RADIOGRAPHY

At photon energies below about 35 keV, attenuation is governed mainly by the atomic number of the tissues because most of the interactions are photoelectric. This property is useful in producing radiographs of materials that do not differ greatly in physical density. Figure 7-4 shows a number of such objects imaged with low-energy x rays.

The use of low-energy x-ray beams presents some technical difficulties. At low voltage the efficiency of x-ray production is reduced (see Chapter 5). Low-energy x rays are rapidly attenuated in tissue, and surface exposures must be relatively high

**FIGURE 7-4**

Low-voltage radiographs: Parrot tulip at 15 kVp, nautilus shell at 40 kVp. (Courtesy of Mathew J. Ottman, University of Minnesota.)

to transmit enough x rays to the radiation receptor on the exit side of the patient. At very low energies (10–15 keV), attenuation of x rays in air is a significant limitation.

Mammography is one of the principal applications of low-energy x rays. In this technique, breast tissues that are similar in physical density, such as glandular tissue and fat, are distinguishable on the basis of their atomic composition. When compared with glandular tissue, for example, fat has a lower effective atomic number because it contains more hydrogen. The dependence of photoelectric absorption on Z^3 permits delineation of these tissue constituents in the image. Microcalcifications in the breast can be seen because of differences in both atomic number and physical density.

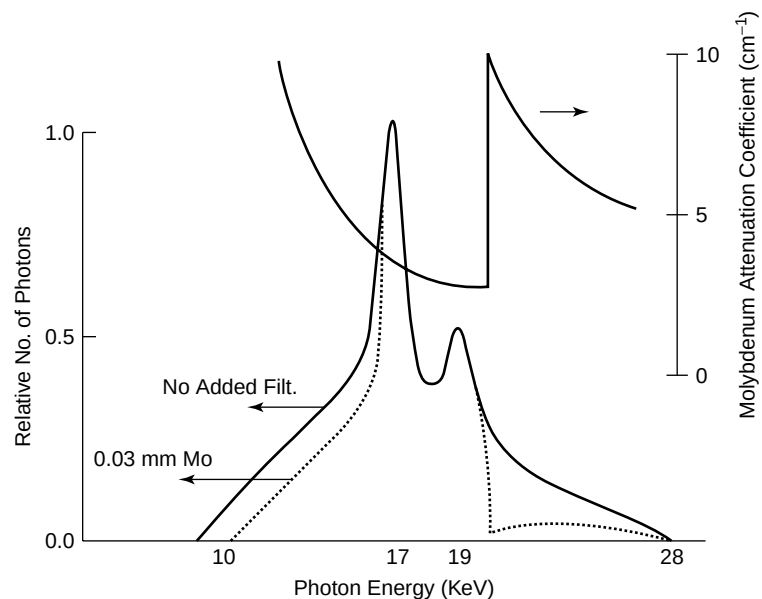
One approach to mammography is to use a tungsten target and relatively low tube voltage (30–45 kVp) to produce a low energy bremsstrahlung x-ray beam. A preferred approach is to use an x-ray target of molybdenum to produce a bremsstrahlung x-ray beam as well as characteristic x rays of approximately 17 and 19 keV. By filtering this x-ray beam with a molybdenum absorber, the very low energy part of the beam is removed so that most of the x rays are above 10 keV. The molybdenum K-absorption edge occurs at 19 keV. X rays with energies above the K-absorption edge are also relatively strongly absorbed. The molybdenum filter is virtually transparent to characteristic x rays from the molybdenum target that fall just below the absorption edge. Hence, most of the characteristic x rays pass through the filter without attenuation (Figure 7-5).

■ CONTRAST MEDIA

Many anatomic structures may be visualized more clearly with x rays if a material is introduced to increase or decrease the x-ray attenuation. Suitable materials are referred to as *contrast agents* or *contrast media*. A few radiographic examinations that may be improved with contrast media are listed in Table 7-3. Many of the agents contain either iodine or barium because the attenuation coefficient for iodine ($Z = 53$) and barium ($Z = 56$) greatly exceeds that for soft tissue ($Z_{\text{eff}} = 7.4$). Consequently, a structure containing an iodinated or barium-containing compound is clearly distinguishable from adjacent soft tissue. The attenuation coefficients for barium and iodine even exceed the coefficient for lead between the K-absorption edges for iodine (33 keV) and lead (88 keV) (Figure 7-6). Within this range of photon energies, x rays are attenuated more rapidly by iodine or barium than by an equal mass of lead. Because most photons in a diagnostic x-ray beam possess an energy between 33 and 88 keV, iodinated and barium-containing compounds are better contrast agents, gram for gram, than would be compounds containing lead or other high- Z elements, even if the toxicity of these compounds did not prohibit their use. Compounds containing iodine or barium are

TABLE 7-3 Contrast Materials for Radiography

Upper GI	Barium
Lower GI	Barium
Small bowel series	Barium
Angiography	Iodine
Urography	Iodine
Myelography	Iodine, air

**FIGURE 7-5**

X-ray spectrum produced by a molybdenum-target x-ray tube operated at 28 kVp (vertical axis on the left) and the attenuation coefficient for molybdenum used to filter the x-ray beam (vertical axis on the right). Larger values of the attenuation coefficient indicate that more photons are attenuated by the filter. The central part of the x-ray spectrum, containing the characteristic peaks at 17 and 19 keV, passes through the filter with little attenuation.

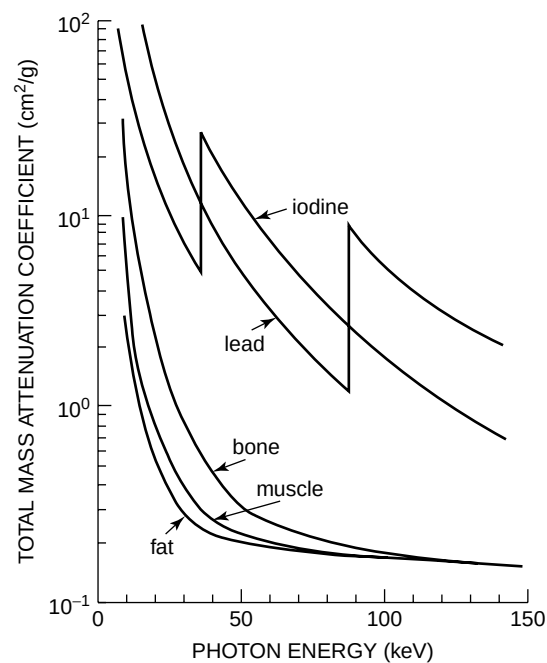
Iodinated contrast agents are water-soluble and, when injected into the circulatory system (angiography), mix with the blood to increase its attenuation relative to surrounding soft tissues. In this manner, blood vessels can be seen that are invisible in x-ray images without a contrast agent.

A thick solution of a barium-containing compound can be introduced into the GI tract by swallowing or enema. The solution outlines the borders of the GI tract to permit visualization of ulcers, polyps, ruptures, and other abnormalities.

Contrast agents have also been developed for use in ultrasound (solutions containing microscopic gas bubbles that reflect sound energy) and magnetic resonance imaging (solutions containing gadolinium that affect the magnetic relaxation of tissues).

relatively nontoxic and may be used in a wide variety of radiographic examinations. One advantage of barium over iodine compounds is their miscibility into solutions of higher physical density.

Air may be introduced into certain locations (e.g., the subarachnoid space of the spinal column) to displace tissues and fluids that interfere with visualization of

**FIGURE 7-6**

Mass attenuation coefficients for fat, muscle, bone, iodine, and lead as a function of photon energy.

anatomic structures of interest. The density of air is very low, and x rays are transmitted through the air-filled cavities with little attenuation. Hence, the introduction of air improves the visibility of structures in or adjacent to air-filled cavities.

PROBLEMS

- 7-1. Referring to Figure 7-1, discuss why x rays generated at low voltage are used to distinguish fat from muscle.
- 7-2. Referring to Figure 7-6, explain why iodine and barium are used in contrast agents.
- 7-3. Why is air an effective contrast agent if the effective atomic number of air ($Z_{\text{eff}} = 7.65$) is near that for muscle ($Z_{\text{eff}} = 7.4$)?
- 7-4. Discuss the advantages and disadvantages of high-voltage radiography.
- 7-5. Plot curves of transmission and absorption for a diagnostic x-ray beam penetrating successive layers of muscle, fat, muscle, bone, and muscle.

SUMMARY

- The f factor relates the radiation exposure (C/kg) in air to the absorbed dose (Gy) in a medium.
- Differences in the following properties among tissues permit the differentiation of various tissues in an x-ray image:
 - Atomic number
 - Electron density (electrons/kg)
 - Physical density (kg/m^3)
- Radiography at higher voltages suppresses shadows cast by bone and is helpful in chest radiography and certain GI studies.
- Radiography at lower voltages enhances soft-tissue differentiation in tissues with low subject contrast.
- Contrast media are often used to enhance contrast of accessible tissues and structures such as the gastrointestinal tract and the cardiovascular system.

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RADIATION DETECTORS FOR QUANTITATIVE MEASUREMENT

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OBJECTIVES

After completing this chapter, the reader should be able to:

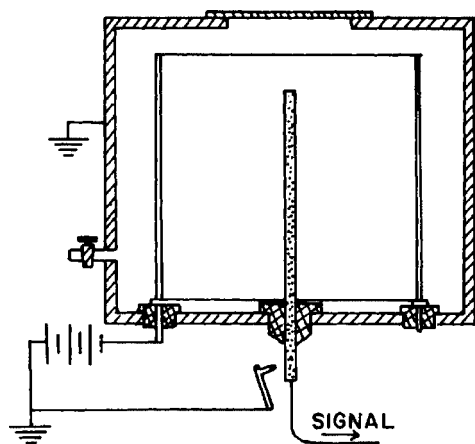
- Compare and contrast pulse-type and current-type ionization chambers.
- Explain the relationship between ionization current and electrode voltage.
- State four uses of ionization chambers in radiology and nuclear medicine.
- Describe the principle of operation of
 - Proportional counters
 - Geiger-müller tubes
 - Solid scintillation detectors
 - Photomultiplier tubes
 - Liquid scintillation detectors
 - Semiconductor detectors

All image receptors used with ionizing radiation are, in a sense, radiation detectors. The properties of image receptors (image intensifiers, computed radiography plates, flat-panel large area detectors, and film-screen systems) are discussed in chapters where the individual imaging techniques are covered. In this chapter we review the principles of operation of radiation detectors that are used to make quantitative measurements of exposure, dose, dose rate, and count rate.

IONIZATION CHAMBERS

Ion pairs are produced as energy is deposited in a medium by ionizing radiation. If a gas is used as the attenuating medium, the ion pairs may be collected by charged electrodes placed in the medium. The ion pairs migrate toward the charged electrodes with a “drift velocity” that depends on the type and pressure of the gas between the electrodes and on the potential difference and distance between the electrodes. In a gas-filled ionization chamber, the voltage between the electrodes is increased until all ion pairs produced by the impinging radiation are collected. However, the voltage remains below that required to produce additional ion pairs as the ion pairs produced by radiation interactions migrate to the collecting electrodes. Consequently, the electrodes receive only ion pairs that result directly from interactions of ionizing radiation with gas in the chamber.

An ionization chamber designed to detect radiation from radioactive sources of low activity consists of parallel-plate or coaxial electrodes in a volume occupied by a filling gas. A parallel-plate chamber resembles the free-air ionization chamber that serves as a calibration standard for ionization chambers used to calibrate x- and γ -ray beams used in radiation therapy.¹ A coaxial chamber is composed of a central electrode in the form of a straight wire or wire loop that is charged positively with respect to the surrounding cylindrical case (Margin Figure 8-1). The entrance of radiation into the chamber results in an electrical current or voltage pulses produced as ion pairs are collected by the electrodes.



MARGIN FIGURE 8-1
A simple coaxial ionization chamber.

Pulse-Type Ionization Chambers

Consider a 1.75-MeV α -particle traversing the collecting volume of an ionization chamber. If the gas in the collecting volume is air, then the α -particle loses an average energy of 33.85 eV for each ion pair (IP) produced. If the kinetic energy of the α -particle is dissipated completely within the collecting volume, then $(1.75 \times 10^6 \text{ eV}) / (33.85 \text{ eV/IP}) = 5 \times 10^4$ ion pairs are produced. Electrons liberated by the radiation migrate rapidly to the central electrode (anode) and reduce the positive charge of this electrode. Usually, electrons are collected within a microsecond after their liberation. The heavier, positively charged ions migrate more slowly toward

the negative case (cathode) of the chamber. As the positive ions approach the case, they induce a negative charge on the case that masks the total reduction in voltage between the electrodes. Hence the total reduction in voltage between the electrodes is not attained until all positive ions within the chamber have been neutralized. Usually, a few hundred microseconds are required to neutralize the positive ions. In most pulse-type ionization chambers, to ensure a more rapid response, only that portion of the reduction in voltage that is created by the collection of electrons is utilized in forming a voltage pulse.

Example 8-1

What size voltage pulse is produced when a 1.75-MeV α -particle is absorbed totally within an air-filled ionization chamber with a capacitance of 10 picofarads? (10 picofarads = 10×10^{-12} farads.)

A 1.75-MeV α -particle produces 5×10^4 IP = 5×10^4 electrons plus 5×10^4 positive ions.

$Q = (5 \times 10^4 \text{ electrons}) (1.6 \times 10^{-19} \text{ coulombs/electron}) = 8 \times 10^{-15} \text{ coulombs}$. The voltage pulse produced is

$$\begin{aligned} V &= \frac{Q}{C}, \quad \text{where } C = 10 \times 10^{-12} \text{ farads} \\ &= \frac{8 \times 10^{-15} \text{ coulombs}}{10 \times 10^{-12} \text{ farads}} \\ &= 8 \times 10^{-4} \text{ V} \\ &= 0.8 \text{ mV} \end{aligned}$$

In a pulse-type chamber designed to produce voltage pulses by the rapid collection of electrons, no interference with the migration of these electrons can be tolerated. Gases such as oxygen, water vapor, and the halogens have an affinity for electrons. These gases should not be present in the collecting volume of an ionization chamber. Other gases such as helium, neon, argon, hydrogen, nitrogen, carbon dioxide, and methane form negative ions only rarely by combining with electrons. These gases may be used as filling gases in ionization chambers.

The average kinetic energy of β -particles (negatrons and positrons) is less than that of most α -particles. Furthermore, the specific ionization is 1/100 to 1/1000 less for negatrons and positrons than for α -particles. Usually, α -particles expend their entire kinetic energy by interacting with the gas in the collecting volume of an ionization chamber. Negatrons and positrons usually strike the wall of the chamber before dissipating all of their kinetic energy. For these reasons, voltage pulses produced as β -particles traverse an ionization chamber are much smaller than those produced by α -particles. Pulses produced by x and γ rays are even smaller. Ionization chambers are usually operated in the pulse mode for the detection of α -particles; other radiations with lower specific ionization are usually measured by operation of the chamber in the current mode.

Current-Type Ionization Chambers

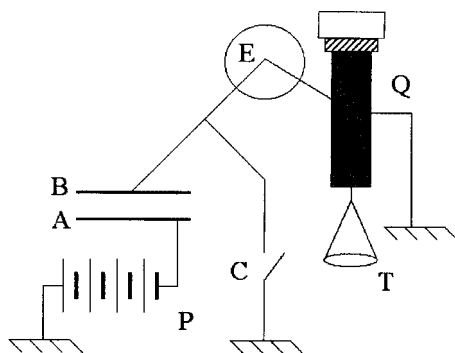
Electrons collected by the anode of an ionization chamber constitute a direct current that may be amplified and measured with a conventional dc meter. In general, this approach is unsatisfactory because instability and zero drift are introduced by the dc amplifier and accurate measurements are difficult to achieve.

Small currents from an ionization chamber may be measured more accurately with an electrometer. An electrometer converts the signal from the ionization chamber into an alternating current that may be amplified with an ac amplifier. An ac amplifier is not subject to the problems of instability and zero drift encountered with a dc



MARGIN FIGURE 8-2

One of the first radiation detectors used was the *gold leaf electroscope*. Two thin strips ("leaves") of gold were suspended at the base of a metal conductor inside of a glass jar. Gold was selected for its electrical conductivity and the ease with which it could be converted into a thin foil. When radiation ionized the air near the outside of the jar, the conductor would acquire an electrostatic charge. This electrostatic charge would distribute itself over the conductor, rendering the two leaves with charge of the same sign. The leaves would then repel each other, causing them to extend outward at some angle. The angle of the leaves was influenced by the amount of electrostatic charge acquired by the conductor, which served as a crude estimate of the amount of ionization in the air outside of the jar.

**MARGIN FIGURE 8-3**

The piezoelectric electrometer/ionization chamber was developed by Pierre Curie in 1897 to measure ionization produced in air due to interactions caused by the emanations from radioactive material. The device was built to aid his wife, Marie Curie, in her efforts to isolate the source of radioactivity from an ore, pitchblende, that was known to fog photographic film and was therefore thought to be “naturally” radioactive. Shown here is a reproduction of a diagram that appeared in Madame Curie’s doctoral dissertation. A battery P produces a voltage across parallel plates A and B when switch C is closed. If a powder of the radioactive ore is placed on one of the plates, ions produced in the air between the plates will produce an electrical current. This small “ionization” current may be measured by determining the voltage needed to offset it. The voltage is produced by adding weights to a pan T to apply force to a piezoelectric quartz crystal Q such that the electrometer E records no net current.

amplifier. The amplified alternating current may then be measured to within $\pm 0.05\%$ precision by one of two methods, referred to as the *voltage-drop method* and the *rate-of-charge method*.

With the voltage-drop method, the amplified alternating current is rectified and directed through a precision resistance. The voltage developed across the resistance is proportional to the current. With the rate-of-charge method, the current is rectified and collected by a precision capacitor. The rate of collection of electrical charge on the plates of the capacitor is proportional to the current. The rate-of-charge method may be used with currents smaller than those measured by the voltage-drop method. The voltage-drop method without current amplification is used in portable survey meters such as the “cutie pie” (Margin Figure 8-4).

Example 8-2*

A 37-Bq sample of $^{14}\text{CO}_2$ is contained within a current-type ionization chamber. The ionization current is converted to alternating current and is measured by the voltage-drop method with a precision resistance of $10^{12} \Omega$. Assuming that all the energy of the β -particles from ^{14}C is deposited in the gas and that the ac signal is not amplified, what voltage is developed across the precision resistance?

$$37 \text{ Bq} = 37 \text{ disintegrations/sec}$$

The average energy of β -particles from ^{14}C is $0.045 \text{ MeV} = 45 \times 10^3 \text{ eV}$. The average energy dissipated per second in the counting volume is

$$(37 \text{ disintegrations/sec})(45 \times 10^3 \text{ eV/disintegration}) = 16.6 \times 10^5 \text{ eV/sec}$$

The number of electrons released per second is

$$\frac{16.6 \times 10^5 \text{ eV/sec}}{33.85 \text{ eV/IP}} = 4.9 \times 10^4 \text{ IP/sec} = 4.9 \times 10^4 \text{ electrons/sec}$$

The charge liberated per second is

$$\begin{aligned} (4.9 \times 10^4 \text{ electrons/sec})(1.6 \times 10^{-19} \text{ coulombs/electron}) \\ = 7.8 \times 10^{-15} \text{ coulombs/sec} \end{aligned}$$

$$1 \text{ coulomb/sec} = 1 \text{ A}$$

$$7.8 \times 10^{-15} \text{ coulombs/sec} = 7.8 \times 10^{-15} \text{ A}$$

The voltage drop V across a resistance is the product of the resistance in ohms and the current in amperes:

$$V = IR$$

$$V = (7.8 \times 10^{-15} \text{ A})(10^{12} \Omega)$$

$$= 7.8 \times 10^{-3} \text{ V}$$

$$= 7.8 \text{ mV}$$

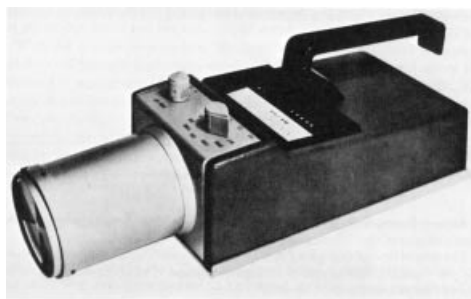
Example 8-3†

Repeat Example 8-2, but find the rate of change in voltage across the plates of a precision 10-picofarad capacitor.

The rate of flow of charge is $7.8 \times 10^{-15} \text{ coulombs/sec}$. Because $V = Q/C$ (where C is the capacitance), the rate of change of voltage is given by $V_r = Q_r/C$

*Modified from data of Chase and Rabinowitch.²

†Modified from data of Chase and Rabinowitch.²

**MARGIN FIGURE 8-4**

A “cutie pie” portable survey meter. The scale of the meter is calibrated to read directly in units of milliroentgens per hour, with a range switch that decreases the sensitivity of the meter by factors of 10.

(where) Q_r is the rate of flow of charge):

$$\begin{aligned}
 V_r &= \frac{7.8 \times 10^{-15} \text{ coulombs/sec}}{10 \times 10^{-12} \text{ farads}} \\
 &= 7.8 \times 10^{-4} \text{ V/sec} \\
 &= (7.8 \times 10^{-4} \text{ V/sec})(60 \text{ sec/min}) \\
 &= 46.8 \times 10^{-3} \text{ V/min} \\
 &= 47 \text{ mV/min}
 \end{aligned}$$

In Margin Figure 8-5, the ionization current is plotted as a function of the voltage applied across the electrodes of an ionization chamber. At low voltages, the electrons and positive ions are not strongly attracted to the electrodes, and some of the ion pairs are lost by recombination. The attraction for ion pairs increases with the voltage between the electrodes, and fewer ion pairs recombine. When the voltage between the electrodes exceeds the *saturation voltage*, the electrodes collect all ion pairs produced by the radiation. The saturation voltage for a particular ionization chamber depends upon the design of the chamber, the shape and spacing of the electrodes, and the type and pressure of the gas in the chamber. No increase in ionization current is observed as the electrode voltage is raised a few hundred volts above saturation because all ion pairs produced by the radiation are collected. This region of voltage is referred to as the *ionization chamber plateau*. The ionization current increases abruptly at the end of the plateau. This increase is due to amplification of the signal caused by the production of additional ion pairs as electrons liberated by the incident radiation gain energy on their way to the anode. Ionization chambers are operated at a voltage below that which causes signal amplification. At any particular voltage, the ionization current produced by an α -emitting sample is much greater than that produced by a sample that emits β -particles. The reduced signal for the β -emitting sample reflects the reduced ionization produced by β -particles.

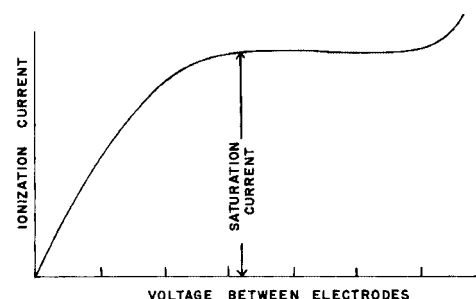
Uses of Ionization Chambers

Radiation from solid, liquid, and gaseous samples may be measured with an ionization chamber and electrometer. The activity of liquid samples prepared for administration to patients often is determined by placing the vial or syringe containing the sample into a well-type ionization chamber referred to as an *isotope calibrator* (Margin Figure 8-6). Gaseous samples may be counted by filling an ionization chamber with the radioactive gas. For example, ionization chambers may be used to measure the amount of $^{14}\text{CO}_2$ in air expired by patients metabolizing compounds labeled with ^{14}C .

Portable survey instruments such as that depicted in Margin Figure 8-4 are frequently used in nuclear medicine to monitor exposure rates in the vicinity of radioactive sources and patients receiving therapeutic quantities of radioactive material. A neutron detector may be constructed by filling an ionization chamber with BF_3 gas or by coating the wall of an ionization chamber with lithium or boron. Ionization is produced within the chamber by α -particles and recoil nuclei that are liberated during interactions of neutrons with the lithium or boron.

PROPORTIONAL COUNTERS

The small signals from an ionization chamber must be amplified greatly before they are measured. Because of the introduction of electronic noise and instability, amplification by electrical circuits is difficult to achieve without distortion of the signal. This problem may be reduced if the signal is amplified within the counting volume of the chamber. If the potential difference between the electrodes of a chamber is raised beyond a certain voltage, electrons liberated by radiation traversing the chamber are



MARGIN FIGURE 8-5

The ionization current from an ionization chamber is plotted as a function of the voltage between electrodes. Recombination of ion pairs occurs at voltages below that furnishing a saturation current. At high voltages, the signal is amplified by ionization produced as electrons are accelerated toward the anode.



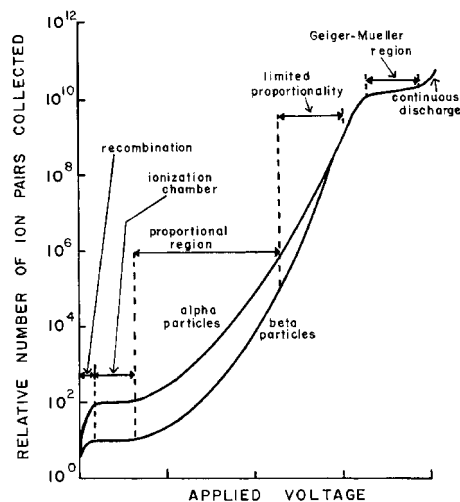
MARGIN FIGURE 8-6

Well-type ionization chamber and electrometer (isotope calibrator) used to measure the activity of a radioactive sample prior to administration of the sample to a patient. The volume of the sample to be administered is computed from the measured activity, often automatically by the isotope calibrator. (Courtesy of RADX Corp.)

Townsend effect

Sir John Sealy Edward Townsend (1868–1957) was a British physicist who was a pioneer in the study of electrical conduction in gases. He did his graduate work in the Cavendish Laboratory under J. J. Thompson, the discoverer of the electron. Townsend made the first direct measurement of the unit electrical charge (e). In 1901 he discovered that gas molecules can be ionized by collision with ions—known today as the “Townsend effect,” a principle that is used in the modern proportional counter.

Michael Faraday (1791–1867) was born into the working class of eighteenth-century England. At the age of 14, he was apprenticed to a bookbinder and made a habit of reading the books he was binding. Through reading, he developed an interest in science. In that time, science was done by the wealthy elite of British society. Faraday attended a lecture given by a prominent physicist, Sir Humphry Davy, took careful notes, and presented Sir Davy with a bound set. He convinced Sir Davy to hire him as an assistant and quickly established a reputation as a scientist in his own right. He eventually derived the law of electromagnetic induction—namely, that a changing magnetic field creates an electrical field that can cause current to flow in an electrical conductor. This principle explains the electrical generator and also the origin of the MR signal that is induced in a receiver coil by precessing magnetic moments.



MARGIN FIGURE 8-7

Relationship between the total number of ion pairs produced in a gas-filled detector and the high voltage between electrodes of the detector.

accelerated to a velocity great enough to produce additional ionization. Most of the additional ionization occurs near the anode of the chamber. As a result, many (10^6 to 10^7) electrons and positive ions are collected by the electrodes for a much smaller number (10^3 to 10^5) of ion pairs produced directly by radiation entering the chamber. This process is referred to as *gas amplification* or the *Townsend effect*. The *amplification factor* for the chamber is the ratio of the total number of ion pairs produced within the chamber to the number liberated directly by radiation entering the chamber. The amplification factor depends on the construction of the chamber and the type of gas enclosed within the chamber. The amplification factor varies from 10^2 to 10^4 for most proportional counters providing a signal (approximately 1 mV) that requires only a small amount of external amplification.

The number of ion pairs produced in a gas-filled chamber is plotted in Margin Figure 8-7 as a function of the voltage applied across the electrodes. The voltage between electrodes must be regulated closely because the amplification factor is affected greatly by small changes in voltage. In the *proportional region*, the amount of charge collected by the electrodes increases with the number of ion pairs produced initially by the impinging radiation. Consequently, the size of the signal from a proportional chamber increases with the amount of ionization produced by radiation that traverses the chamber.

Example 8-4*

An α -particle produces 10^5 ion pairs (IP) in a proportional chamber with an amplification factor of 10^3 . How many IPs are collected by the electrode?

With an amplification factor of 10^3 , 1000 IPs are collected for every ion pair liberated by the incident radiation. The total number of IPs collected is

$$(10^5 \text{ IPs})(10^3) = 10^8 \text{ IPs}$$

Example 8-5*

Repeat the calculation in Example 8-4 for a β -particle that produces 10^3 IPs within the chamber.

For an amplification factor of 10^3 , we have

$$(10^3 \text{ IPs})(10^3) = 10^6 \text{ IPs}$$

Because of its rapid response to ionizing events, the multiwire proportional chamber has been investigated as an imaging device for nuclear medicine.^{3,4} By raising the pressure of the counting gas to 10 atm or more, the sensitivity of the chamber to γ rays can be improved. The γ -ray sensitivity can also be improved by placing high-Z foils in front of the chamber to convert incoming photons to photoelectrons and Compton electrons. Although the intrinsic resolution of multiwire proportional chambers is excellent, the resolution achievable in practice has been no better than that for other nuclear medicine imaging devices.

At voltages higher than the proportional region, α -particles and other densely ionizing radiations initiate ionization of most of the atoms of gas in the vicinity of the anode. If the chamber is operated at a voltage in this region, then the number of ion pairs collected is not strictly proportional to the ionization produced directly the radiation. Hence this voltage region is referred to as the *region of limited proportionality*. Proportional chambers are not operated routinely in the region of limited proportionality.

■ GEIGER-MÜLLER TUBES

If the potential difference between the electrodes of a gas-filled detector exceeds the region of limited proportionality (Margin Figure 8-7), then the interaction of a

charged particle or x or γ ray within the chamber initiates an *avalanche of ionization*, which represents almost complete ionization of the counting gas in the vicinity of the anode. Because of this avalanche process, the number of ion pairs collected by the electrodes is independent of the amount of ionization produced directly by the impinging radiation. Hence the voltage pulses (usually 1 to 10 V) emerging from the detector are similar in size and independent of the type of radiation that initiates the signal. The range of voltage over which signals from the detector are independent of the type of radiation entering the detector is referred to as the *Geiger-Müller (G-M) region*. For detectors operating in this voltage region, the amplification factor is 10^6 to 10^8 .

In Margin Figure 8-8, the number of pulses (or counts) recorded per minute is plotted as a function of the voltage across the electrodes of a G-M detector exposed to a radioactive source. No counts are recorded if the voltage is less than the starting voltage because voltage pulses formed by the detector are too small to pass the discriminator and enter the scaler. As the voltage is raised slightly above the starting voltage, some of the pulses are transmitted by the discriminator and recorded. At the *plateau threshold voltage*, all pulses are transmitted to the scaler. Increasing the voltage beyond the plateau threshold does not increase the count rate significantly. Consequently, relatively inexpensive high-voltage supplies that are not exceptionally stable may be used with a G-M detector. G-M detectors usually are operated at a voltage about one-third of the way up the plateau. In Margin Figure 8-8, for example, an operating voltage of 1150 V might be selected for the detector. The voltage range encompassed by the plateau varies with the construction of the G-M detector and with the counting gas used.

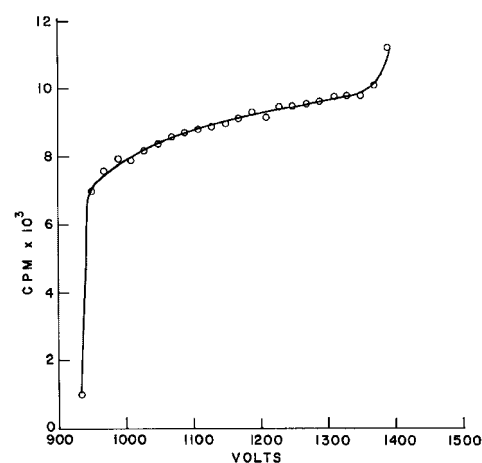
Atoms of counting gas near the anode may be ionized spontaneously if the voltage applied to the detector is raised beyond the G-M plateau. This region of voltage is referred to as the *region of spontaneous discharge* or *region of continuous discharge* because the counting gas may be ionized in the absence of radiation. A G-M tube may be permanently damaged by the application of voltages higher than the G-M plateau.

A few G-M detectors are illustrated in Margin Figure 8-9. The anode is a thin wire of tungsten or stainless steel in the center of the detector. The anode is surrounded by a metal or glass cathode that is coated internally with a conducting layer of graphite or evaporated metal. The efficiency of a G-M detector for high-energy x and γ rays may be increased by coating the cathode with a heavy metal such as bismuth or lead. The end window G-M detector (Margin Figure 8-9A) is often used for assay of radioactive samples. The window is usually constructed from split mica with a thickness of a few milligrams per square centimeter. The thin window admits α -particles and low-energy β -particles into the counting volume. Ultrathin Mylar windows between 100 and 200 $\mu\text{g}/\text{cm}^2$ thick are available, but they must be used with a flow counter and a continuous supply of counting gas because windows this thin are permeable to the counting gas.

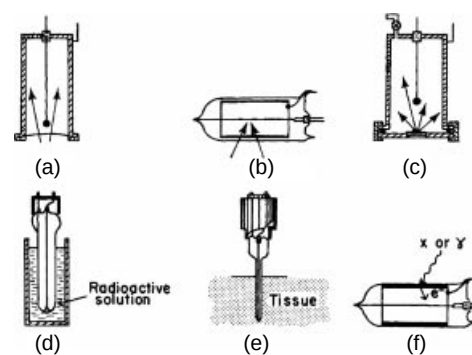
Detectors with thin walls are used primarily with portable survey meters (Margin Figure 8-9B). The walls are metal or glass and are about 30 mg/cm^2 thick. With the windowless flow detector (Margin Figure 8-9C), the radioactive sample is sealed inside the chamber while it is counted. The thin-wall dipping tube (Margin Figure 8-9D) may be inserted into a radioactive solution. With the needle probe detector (Margin Figure 8-9E), the sensitive volume occupies the tip of a long, thin probe. These detectors may be used to locate radioactive material concentrated in tissues within the body. The chamber with a high-Z wall (Margin Figure 8-9F) is used primarily for the detection of x and γ rays.

When an ionizing event is initiated in a G-M detector, an avalanche of electrons is created along the entire length of the anode. The residual positive ions require 200 μs or longer to migrate to the cathode. During the time required for migration of the positive ions, the detector will not respond fully to additional radiation that enters the counting volume. The curve in Margin Figure 8-10 depicts the response of the detector as a function of time after an ionizing event. During the “dead time,” the detector is completely unresponsive to additional radiation. An ionizing event occurring within the “recovery time” produces a voltage pulse that is smaller than

C. T. R. Wilson, a British Scientist, invented the *cloud chamber* in 1912. A cloud chamber is a detector that allows researchers not only to detect the presence of ionizing photons and particulate radiation but also to determine the path taken by the particles through the chamber, even if the path is curvilinear or spiral. He accomplished this feat by preparing a container in which a gas was “supersaturated”—that is, at the appropriate temperature and pressure to condense into a liquid, but still existing in a gaseous state due to the absence of nucleation sites. When ionizing radiation enters a cloud chamber detector, the formation of ion pairs produces nucleation sites that cause a stream of condensation to appear along the path of the particle. A similar phenomenon occurs when a jet plane disturbs supersaturated layers of the upper atmosphere, leaving a “jet stream” pattern of condensation along its path.



MARGIN FIGURE 8-8
Characteristic curve for a G-M detector.



MARGIN FIGURE 8-9
Various G-M detectors. A: End window. B: Side window. C: Windowless flow. D: Thin-wall dipping tube. E: Needle probe. F: Heavy-metal wall.

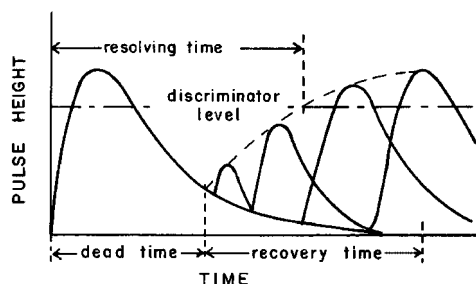
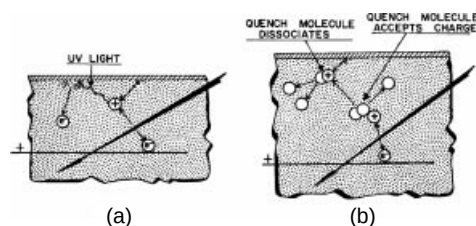
**MARGIN FIGURE 8-10**

Diagram illustrating the formation of a voltage pulse in a G-M detector as a function of time after an ionizing event.

**MARGIN FIGURE 8-11**

A: Self-perpetuating discharge of a G-M detector caused by bombardment of the cathode by ultraviolet and x-ray photons that are released as positive ions are neutralized. **B:** Molecules of quench gas accept the charge of positive ions and dissociate when neutralized near the cathode. In this way the self-perpetuating discharge is prevented.

The slope of the plateau may be computed as follows:

$$\text{Slope (\%/100 V)} = \frac{2[(\text{cpm})_2 - (\text{cpm})_1]10^4}{[(\text{cpm})_2 + (\text{cpm})_1](V_2 - V_1)}$$

where $(\text{cpm})_2$ is the count rate at voltage V_2 on the plateau and $(\text{cpm})_1$ is the count rate at voltage V_1 on the plateau.

**MARGIN FIGURE 8-12**

Portable survey meter equipped with an end window G-M tube.

normal. The “resolving time” is the time between an ionizing event and second event that furnishes a pulse large enough to pass the discriminator and be counted.

Positive ions that approach the cathode of a G-M detector dislodge electrons from the wall of the chamber. As these electrons combine with the positive ions, ultraviolet and x-ray photons are released. Some of these photons strike the chamber wall and release electrons that cause the chamber to remain discharged. If this secondary release of electrons is permitted to occur, the detector will be unresponsive to radiation after the first ionizing event. Self-perpetuating discharge of a G-M detector is diagramed in Margin Figure 8-11A.

Various methods have been devised to “quench” the self-perpetuating discharge of G-M detectors. The most common method is to add a small concentration (about 0.1%) of a selected gas to the counting gas. The gases used most often as “internal quench agents” are polyatomic organic gases (e.g., amyl acetate or ethyl alcohol vapor) or halogens (e.g., Br_2 or Cl_2). Their effect is shown in Margin Figure 8-11B. As the positive ions move toward the cathode, they collide with and transfer charge to molecules of quench gas. The charged molecules of quench gas migrate to the cathode and dislodge electrons from the chamber wall. Energy released as the dislodged electrons combine with the charged molecules causes the dissociation of molecules of the quench gas. The dissociation is irreversible with a polyatomic organic gas, and the useful life is 10^8 to 10^{10} pulses for a G-M detector quenched with one of these agents. Halogen-quenched tubes have an infinite useful life, theoretically, because the molecules recombine after dissociation.

The counting gas used routinely in G-M detectors is an inert gas such as argon, helium, or neon. “Geiger gas” used in flow counters is composed of 99% helium and about 1% butane or isobutane. For G-M tubes quenched with an organic gas, the plateau should be 200 to 300 V long and should have a slope not greater than 1% to 2%/100 V. Halogen-quenched tubes have a shorter plateau (100 to 200 V) and a plateau slope of 3% to 4%/100 V.

The detection efficiency of a G-M counter is about 1% for x and γ rays and nearly 100% for α - and β -particles that enter the counting volume. Of course, many α and low-energy β -particles are absorbed by the window of the detector. Windowless flow counters are often used to detect these particles. Shown in Margin Figure 8-12 is a survey meter equipped with an end-window G-M tube for detecting the presence of radioactive contamination.

SOLID SCINTILLATION DETECTORS

Gas-filled chambers are not efficient detectors for x- and γ -ray photons because these radiations pass through the low-density gas without interacting. The probability of x- and γ -ray interaction is increased if a solid detector with a high density and atomic number is used. Atoms of a solid are immobile, however, and an interaction cannot be registered by the collection of electrons and positive ions. Instead, the interaction must be detected by some alternate method. In a scintillation crystal, light is released as radiation is absorbed. The light impinges upon a photosensitive surface in a photomultiplier tube. Electrons released from this surface constitute an electrical signal. Scintillation detectors may be used to detect particulate radiation as well as x- and γ -ray photons. For example, liquid scintillators often are used to detect low-energy β -particles.

Principles of Scintillation Detection

When an x or γ ray interacts within a scintillation crystal, electrons are raised from one energy state to a state of higher energy. The number of electrons raised to a higher energy level depends upon the energy deposited in the crystal by the incident x or γ ray. Light is released as these electrons return almost instantaneously to the lower

energy state. In most scintillation detectors, about 20 to 30 photons of light are released for every keV of energy absorbed. The photons of light are transmitted through the transparent crystal and directed upon the photosensitive cathode (photocathode) of a photomultiplier tube. If the wavelength of light striking the photocathode matches the spectral sensitivity of this photosensitive surface, then electrons are ejected. The number of electrons is multiplied by various stages (dynodes) of the photomultiplier tube, and a signal is provided at the photomultiplier anode that may be amplified electronically and counted. The size of the signal at the anode is proportional to the energy dissipated in the detector by the incident radiation.

Scintillation Crystals

Gamma rays from radioactive samples are often detected with a scintillation crystal. Alkali halide crystals usually are used because the probability of photoelectric interactions is increased by the presence of the high- Z halide component. Sodium iodide is the alkali halide used most frequently, although crystals of cesium iodide and potassium iodide are available at higher cost. Crystals of sodium iodide up to 9 in. in diameter by 9 in. thick or 20 in. in diameter by 0.5 in. thick are available commercially. Smaller crystals (e.g., 2 in. in diameter by 2 in. thick) are used routinely for the assay of γ -emitting samples. The efficiency of a crystal of detecting x and γ rays increases with the size of the crystal.

To be used as a scintillation detector, an alkali halide crystal must be “activated” with an impurity. The impurity is usually thallium iodide at a concentration of about 0.1%, and the crystals are denoted as NaI(Tl), CsI(Tl), or KI(Tl).

Highly purified organic crystals (e.g., anthracene and *trans*-stilbene) are used to detect β -particles. The atomic number of these crystals is relatively low, and the probability is reduced that β -particles will be scattered out of the detector after only part of their energy has been dissipated. The sensitivity of an anthracene or *trans*-stilbene detector to γ rays is low, particularly if the crystal is thin. Consequently, β -particles may be detected with limited interference from γ rays.

Mounting Scintillation Crystals

Sodium iodide crystals are hygroscopic and must be protected from moisture. If exposed to moisture, NaI(Tl) crystal turns yellow and absorbs much of the radiation-induced fluorescence so that less light is emitted (reducing the signal available for detection of radiation). The yellow color reflects the release of free iodine. Crystals are mounted in a dry atmosphere and sealed to prevent the entrance of moisture. A light pipe of Lucite, clear glass, or quartz is sometimes attached to the side of the crystal nearest the photocathode. Other surfaces of the crystal are coated with a light-reflecting material (e.g., Al_2O_3 , MgO, or aluminum foil). The crystal may be enclosed within an aluminum canister, perhaps 1/32 in. thick. The canister prevents moisture from reaching the crystal and ambient light from reaching the photocathode. The crystal or light pipe is coupled to the glass face of the photomultiplier tube with a transparent viscous medium such as silicone fluid.

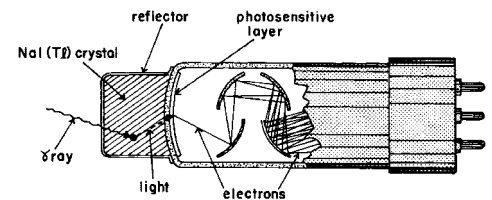
Photomultiplier Tubes

A photomultiplier tube is diagramed in Margin Figure 8-14. The photocathode is usually an alloy of cesium and antimony, often mixed with sodium and potassium (i.e., a *bialkali* photocathode), from which an acceptable number of electrons are released per light photon absorbed. The spectral sensitivity of the alloy must match the wavelength of light emerging from the crystal. The spectral sensitivity of a photocathode with an “S-11 response” is compared in Margin Figure 8-15 with the emission spectrum of light from irradiated NaI(Tl). Only 10% to 30% of the light photons that strike the photocathode cause the ejection of electrons. These electrons are



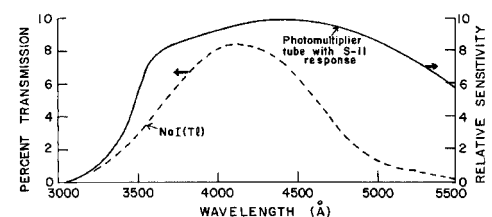
MARGIN FIGURE 8-13

The Sudbury Neutrino Observatory (SNO) is located 6800 feet below ground in the Creighton nickel mine near Sudbury, Ontario. SNO captures the elusive neutrino in a tank filled with 10,000 metric tons of heavy water, water in which each ordinary hydrogen atom (hydrogen with a nucleus consisting of a single proton) is replaced by a deuteron, a hydrogen isotope with a nucleus containing a proton, and a neutron. Note the scale of the detector compared to the workers below it. (Photo from the *New York Times*, June 22, 1999.)



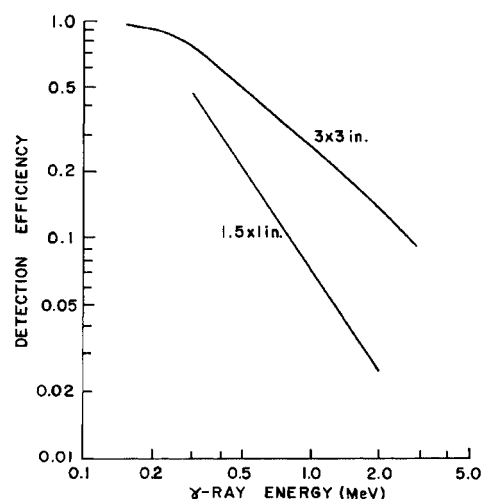
MARGIN FIGURE 8-14

A sodium iodide crystal and photomultiplier tube.



MARGIN FIGURE 8-15

The emission spectrum of NaI(Tl) is matched closely to the spectral sensitivity of a photomultiplier tube with an S-11 response.

**MARGIN FIGURE 8-16**

The detection efficiency of 3×3 -in. and 1.5×1 -in. NaI(Tl) crystals is plotted as a function of the energy of incident γ -rays. The radioactive sources were positioned 7 cm from the 1.5×1 -in. crystal and 9.3 cm from the 3×3 -in. crystal. (From Lazar, N., Davis, R., and Bell, P. *Nucleonics* 1956; 14:52. Used with permission.)

accelerated to the first dynode, a positively charged electrode positioned a short distance from the photocathode. For each electron absorbed by the first dynode, three or four electrons are ejected and accelerated to the second dynode, where more electrons are released. Because photomultiplier tubes contain 6 to 14 dynodes with a potential difference of 100 to 500 V between successive dynodes, 10^6 to 10^8 electrons reach the anode for each electron liberated from the photocathode. The amplification of the signal is very dependent upon the potential difference between dynodes, and the high-voltage supply for dynodes must be very stable.

Electrons collected by the anode are converted to a voltage pulse with an amplitude of a few millivolts to a few volts. This voltage pulse is delivered to the preamplifier, which often is mounted on the photomultiplier tube.

Energy Dependence of NaI(Tl) Detectors

The detection efficiency of a NaI(Tl) scintillation detector decreases with increasing energy of impinging γ rays (Margin Figure 8-16). The efficiency of a scintillation detector for detection of γ rays may be improved by using a larger crystal and by improving the “counting geometry.” For example, a crystal into which the radioactive sample may be inserted (i.e., a well detector) may furnish a detection efficiency greater than that provided by a crystal that receives at best no more than half the γ rays from a radioactive source.

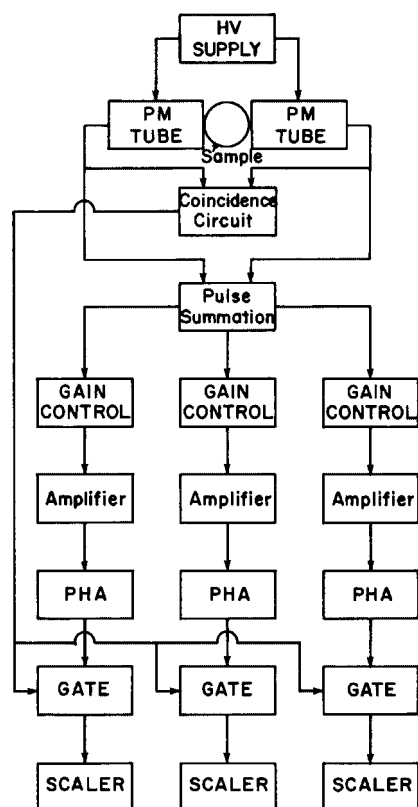
LIQUID SCINTILLATION DETECTORS

With a solid scintillation detector such as a NaI(Tl) crystal, the radioactive sample is positioned outside the detector. In liquid scintillation counting, the radioactive sample is mixed intimately with the scintillating material, and attenuation by materials between the radioactive sample and the scintillating material is reduced to a minimum.^{5,6} Consequently, the detection efficiency is high for radiations with very short range, including weak β -particles such as those from ^3H ($E_{\text{max}} = 0.018$ MeV), ^{14}C ($E_{\text{max}} = 0.156$ MeV), and ^{35}S ($E_{\text{max}} = 0.168$ MeV). Usually light from the mixture of scintillator and radioactive sample is directed toward at least two photomultiplier tubes. The signal from each photomultiplier tube is transmitted by a preamplifier and amplifier to a coincidence circuit (Margin Figure 8-17). The coincidence circuit transmits a voltage pulse to the scaler only if a pulse is received simultaneously from both photomultiplier tubes.

Except for very low energy particles and photons, radiation emitted by the sample usually produces a signal in each photomultiplier tube, and a pulse passes to the scaler for most disintegrations of the sample. However, spurious pulses generated by “thermal noise” in the photomultiplier tubes or preamplifiers are received by the coincidence circuit from one direction only, and a pulse is not passed to the scaler. In this manner, the coincidence circuit reduces the background count rate. Without this circuit, the background count rate would be intolerably high. In older liquid scintillation counters, the scintillation mixture, photomultiplier tubes, and preamplifiers were cooled to a few degrees above zero to help eliminate spurious counts. Newer photomultiplier tubes do not require refrigeration.

The scintillating solution, or “cocktail,” consists of the radioactive sample, solvent, primary fluor, or solute and, if necessary, a secondary fluor. Solvents dissolve the sample and transfer energy from the sites of interaction of the radiation to the molecules of fluor present at a low concentration (1% or less) in the scintillation mixture.

Molecules of the primary fluor release light upon receipt of energy from the solvent molecules. The wavelengths of light emitted by the scintillation cocktail must correspond to the spectral sensitivity of the photocathodes of the photomultiplier

**MARGIN FIGURE 8-17**

A liquid scintillation counter with two photomultiplier tubes and a coincidence circuit to reduce the background count rate.

tubes. Some photomultiplier tubes with quartz windows are sensitive to the light emitted by the primary fluor. With other photomultiplier tubes, however, a secondary fluor must be added to the scintillation solution. The secondary fluor is termed a *wavelength shifter* because light of longer wavelength is emitted by the scintillation solution when the secondary fluor is present.

In a liquid scintillation counter, the size of the voltage pulse depends on the energy dissipated in the scintillation cocktail by a photon or particle emitted by the radioactive sample. The number of pulses of different sizes is shown in Margin Figure 8-18 for β -emitting samples of ^3H and ^{14}C . By adjusting upper and lower discriminators, one isotope may be counted in the presence of the other. For a number of reasons, these spectra do not exactly correspond to the energy distribution of the emitted particles.

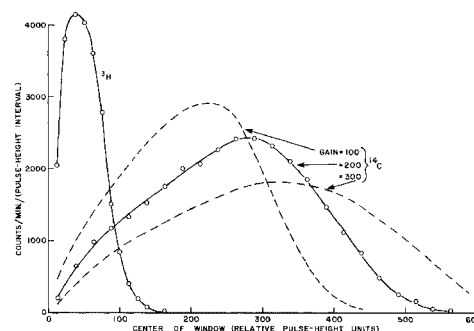
Interference with the production or transmission of light in a liquid scintillation solution is termed *quenching*. Quenching is always present in liquid scintillation counting and is caused by the following:

1. Interference with the mechanism of energy transfer contributed by the sample or other components of the cocktail: *chemical quenching*.
2. Absorption of light by colored materials in the sample: *color quenching*.
3. Passive interference with the mechanism of energy transfer resulting from dilution of the scintillation mixture by the sample or other material: *dilution quenching*.
4. Absorption of light by the scintillation vial, fingerprints on the vial, and so on: *optical quenching*.

Quenching shifts the spectrum for any isotope toward pulses of smaller size. Shown in Margin Figure 8-19 are spectra for a ^{14}C -labeled sample dissolved in a scintillation cocktail and quenched with different amounts of carbon tetrachloride, a chemical quench agent.

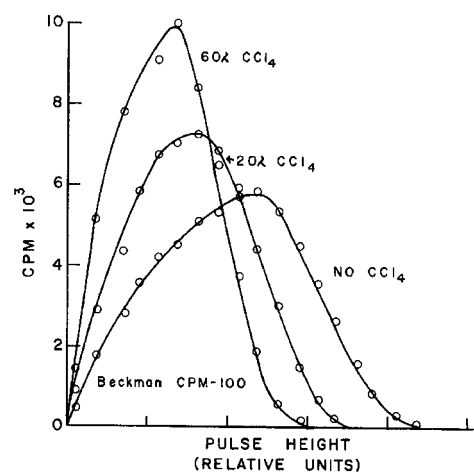
The count rate for a particular sample must be corrected for quenching before the disintegration rate of the sample can be determined. Three methods for quench correction have been developed.⁷ With the *internal spike method* (internal standard method), a cocktail is counted before and after a small quantity of “unquenched” material has been added to the sample. The unquenched material and the sample are labeled with the same radioactive isotope. The count rate without the unquenched material is subtracted from the count rate with the unquenched material. The counting efficiency is the difference in count rate divided by the disintegration rate for the unquenched material. With the *channels ratio method* for quench correction, a sample is counted in two separate “counting windows,” or “channels,” which are defined by upper, lower, and intermediate discriminators. The ratio of the count rates in the two channels varies with the amount of quenching in the cocktail. By reference to a calibration curve of counting efficiency versus the ratio of count rates, the counting efficiency for a particular sample may be determined. With the *external standard method* for quench correction, the sample is counted before and after a γ -emitting source (e.g., ^{137}Cs , radium, or ^{133}Ba) has been positioned adjacent to the scintillation vial. The increase in count rate obtained with the source near the vial varies with the amount of quenching in the sample. The counting efficiency is determined by referring to a calibration curve of counting efficiency versus the ratio of count rates before and after the γ -emitting source has been positioned adjacent to the scintillation vial.

Samples that are not soluble in a liquid scintillation cocktail may be counted by *suspension counting*. Gelling agents such as aluminum stearate, Cab-O-Sil, and Thixcin furnish suspensions of radioactive samples in various counting solutions. Techniques have been developed for counting insoluble samples such as filter paper, paper chromatograms, and Millipore filter disks. Scintillating beads are sometimes used when liquid or gaseous samples are counted by liquid scintillation.



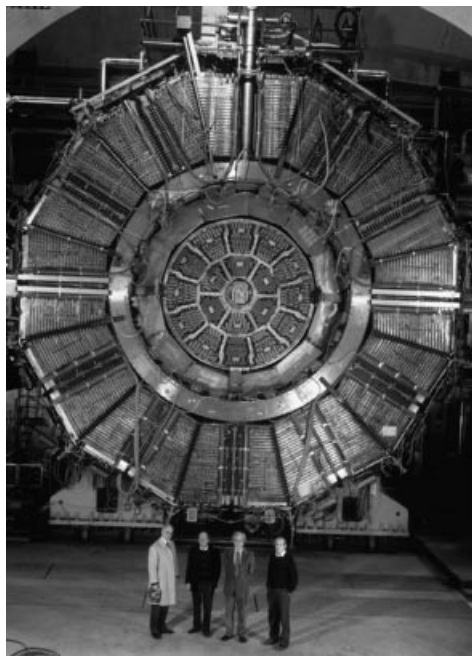
MARGIN FIGURE 8-18

Liquid scintillation spectra for β -particles from ^3H ($E_{\text{max}} = 0.018 \text{ MeV}$) and ^{14}C ($E_{\text{max}} = 0.156 \text{ MeV}$). The dashed curves illustrate the effects of the amplifier gain upon the ^{14}C spectrum.



MARGIN FIGURE 8-19

Effects of quenching on the liquid scintillation spectrum for ^{14}C . The quenching agent is CCl_4 .

**MARGIN FIGURE 8-20**

ALEPH is a particle detector that was built and is operated by a collaboration of several hundred physicists and engineers from 32 universities and national laboratories from around the world. Its purpose is to explore the Standard Model of particle physics and search for manifestations of new physics through observation of particles produced by the large electron–positron collider at the CERN laboratory in Geneva, Switzerland. The detector includes eight separate subsystems, making use of calorimetry, solid state, and other types of detectors. It is surrounded by one of the largest superconducting magnets in the world. Shown here is a photo of the front of the detector.

In an intrinsic semiconductor, such as germanium, electrons are bound to atoms via covalent bonds in specific positions within a crystal structure. If an electron is freed from an atom, it can participate in current conduction by moving through the material. The atom from which the conduction electron originated has a deficiency that may be filled by neighboring electrons, causing the site of the deficiency to move. This moving positively charged “deficiency” region may be regarded as a fictitious particle called a “hole.” At room temperature, intrinsic semiconductors rarely develop electron–hole pairs.

SEMICONDUCTOR RADIATION DETECTORS

Semiconductor detectors are often used for the detection of charged particles and photons emitted by radioactive nuclei. Semiconductor detectors exhibit many desirable properties, including (1) a response that varies linearly with the energy deposited in the detector and does not depend upon the type of radiation that deposits the energy, (2) a negligible absorption of energy in the entrance window of the detector, (3) excellent energy resolution, (4) the formation of pulses with fast rise times, and (5) small detector size.

The mechanism of response of a semiconductor detector resembles that for an ionization chamber. Ionization produced within the sensitive volume of the detector is converted to a voltage pulse that is amplified and counted. The size of the voltage pulse is proportional to the energy expended in the detector by the incident radiation. When compared with an ionization chamber, the voltage pulse is larger and more accurately reflects the energy deposited in the detector. Also, the rise time of the pulse is shorter because the ionization is collected more rapidly.

In most gases, an average energy of 30 to 40 eV is expended per ion pair produced. An ion pair is produced in a silicon semiconductor detector for each 3.5 eV deposited by incident radiation; in a germanium detector, only 2.9 eV is required to produce an ion pair. When compared with an ionization chamber, therefore, many more ion pairs are produced in a semiconductor detector for a given amount of energy absorbed.

Response of Semiconductor Detectors

A semiconductor radiation detector is similar to a transistor and is diagrammed in Margin Figure 8-21. The *p*-type region is composed of a semiconducting element (e.g., germanium or silicon) “doped” with an *electron acceptor impurity* with fewer valence electrons. For example, a *p*-type semiconductor may be obtained by doping tetravalent germanium with trivalent boron, indium, or gallium. The *n*-type region is composed of germanium doped with an *electron donor impurity* such as antimony or lithium. Electrons flow from the *n*-type region to the *p*-type region and establish an electric field across the junction between the two regions. The region in the vicinity of the junction is termed the *depletion region* and may be increased in width by applying a *reverse bias* across the junction (positive potential to the *n*-type region, negative potential to the *p*-type region). The width of the depletion region may also be increased by a process known as *lithium drifting* to produce a Ge(Li) or Si(Li) detector.

If a charged particle or an x or γ ray loses energy within the depletion region, electrons are raised from the valence band to the conduction band, where they can migrate to the positive electrode (the *n*-type region). In the valence band, electrons move closer to the positive terminal by jumping to holes left by the released electrons. Other electrons fill the holes left by the jumping electrons. In this manner, holes migrate toward the negative terminal as if they were positively charged particles. The migration of positive holes in the valence band constitutes a current similar to that provided by electrons moving in the conduction band to the positive terminal. In fact, electrons released from the valence band, together with the positive holes left behind, constitute the ion pairs for a semiconductor detector.

If a potential difference is applied across a pure semiconductor, a current is produced even if the semiconductor is not exposed to ionizing radiation. This current is the sum of (1) a *bulk current* that is dependent on the resistance of the semiconductor and the number of electron–hole pairs produced by thermal excitation and (2) a current caused by *charge leakage* at the surface of the semiconductor. These currents interfere with the identification of signals produced as radiation interacts within the detector. The bulk current is reduced with the *p–n* junction described above, and this barrier to current flow is required in a semiconductor radiation detector. The *p–n*

junction reduces the bulk current in silicon to an acceptable level at room temperature. Even with a p - n junction, however, the bulk current in a germanium detector is too great at room temperature. Consequently, germanium semiconductor detectors must be operated at reduced temperature. Germanium detectors are usually mounted in a cryostat and maintained at the temperature of liquid nitrogen (-190°C). The surface-leakage current is reduced in germanium and silicon by special techniques for constructing the detectors.

The size V of a voltage pulse from a semiconductor detector equals the charge Q collected by the electrodes divided by the capacitance C of the depletion region. The charge Q equals Ne , where N is the number of electron-hole pairs produced and e is the charge of the electron. That is, the size of the voltage pulse is proportional to the energy lost in the detector by the incident radiation. The size of the pulse is not dependent on the specific ionization of the radiation, because the ion pairs are swept away immediately and cannot recombine. Consequently, the response of the detector depends on the energy deposited in the detector but not on the type of radiation that deposits the energy.

The energy required to produce an ion pair in a semiconductor detector is only one-tenth of the energy required in a gas, and the voltage pulse from a semiconductor detector is about 10 times larger than the pulse from a gas-filled ionization detector. For example, a 1-MeV α -particle absorbed completely in the depletion region of a silicon semiconductor detector produces about $[10^6 \text{ eV}/(3.5 \text{ eV/IP})] = 3 \times 10^5$ ion pairs. The same particle produces only about $[10^6 \text{ eV}/(33.85 \text{ eV/IP})] = 3 \times 10^4$ ion pairs in an air-filled ionization chamber. The estimated percent standard deviation $\% \sigma/N$ (Chapter 11) for the pulse is

$$\frac{\% \sigma}{N} = \frac{100}{\sqrt{\text{Pulse size}}}$$

For the semiconductor detector, $\% \sigma/N$ is 0.18 for the pulse produced by a 1-MeV α particle. For the gas-filled detector, $\% \sigma/N$ is 0.58. Consequently, the range of pulse heights produced by the absorption of a given amount of energy is much narrower for a semiconductor detector than for a gas-filled ionization chamber, and the energy resolution of the semiconductor detector is much better. A similar analysis is applicable to the comparison of a semiconductor detector to a scintillation detector. Often, the maximum resolution obtainable with a semiconductor detector is limited by the preamplifier rather than by the detector.

Properties of Semiconductor Detectors

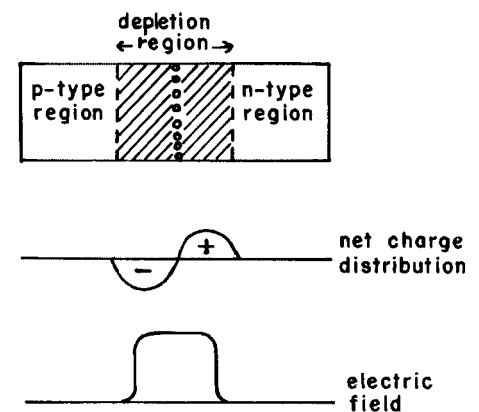
The efficiency of semiconductor detectors is nearly 100% for particulate radiations and relatively high for low-energy x and γ rays. The efficiency for detecting high-energy x and γ rays is lower because the depletion regions of the detectors are small. The atomic number is greater for germanium ($Z = 32$) than for silicon ($Z = 14$), and the γ -detection efficiency is higher.

Applications of Semiconductor Detectors

Because of their excellent energy resolution, semiconductor detectors are used widely for x - and γ -ray spectrometry and for similar laboratory measurements. These detectors have also been used for fluorescence scanning, where quantitative estimates of iodine in the thyroid are obtained by measurement of iodine x rays released as the thyroid is exposed to low-energy x or γ radiation.⁸ Some effort has been directed toward extension of semiconductor detectors to imaging applications in nuclear medicine. Although not in widespread use at the present time, one vendor offers a photodiode coupled to each of 4096 CsI(Tl) scintillation crystals, thereby eliminating the light guide and photomultiplier tubes (see Chapter 12) and reducing the size and weight of the scanner.⁹

However, if impurities, called doping agents, are introduced, a higher level of conduction is produced. In the doped state, light, heat, or ionizing radiation may trigger formation of electron-hole pairs. For example, if an atom of arsenic, having five outer electrons, is introduced into the crystal structure of germanium, which has four outer electrons, four of the arsenic electrons are taken up in covalent bonds while the fifth is free to move about as a negative charge carrier. Therefore, germanium doped with arsenic is an n -type semiconductor. Less than one arsenic atom per million germanium atoms is required to achieve this effect.

Similarly, if boron atoms, which have three outer-shell electrons, are used as a doping agent in germanium, holes appear, yielding a p -type semiconductor.



MARGIN FIGURE 8-21

Diagram of a p - n junction semiconductor detector.

The capacitance C of a semiconductor detector may be computed with the expression:

$$C = 1.44 \frac{A}{W}$$

where A is the area of the junction and W is the width of the depletion region.

PROBLEMS

- *8-1. For a G–M detector operated with a voltage on the G–M plateau:
- Is the pulse produced by an α -particle larger than a pulse produced by a β -particle?
 - Do the size and shape of the pulse vary with the length of the anode? With the diameter of the detector?
- *8-2. Should a G–M detector be used to measure the exposure rate in the vicinity of a storage safe for radioactive nuclides?
- 8-3. What type of radiation detector would you recommend for the following:
- Detection of γ rays from ^{131}I in a patient's thyroid?
 - Detector of β -particles from $[^3\text{H}]\text{thymidine}$ dissolved in water?
 - Detection of β -particles from ^{14}C in a gaseous sample of CO_2 ?
 - Detection of α -particles from a plated source of ^{210}Po ?
 - Detection of radioactive contamination on a workbench in a laboratory where ^{32}P is used? In a laboratory where ^3H is used?
 - Measurement of the exposure rate in the vicinity of a brachytherapy patient?
- ^{131}I in aqueous solution with ^{32}P ?
 - A mixture of ^{51}Cr and ^{131}I in blood?
- *8-4. A γ ray from ^{241}Am (60 keV) is absorbed completely in an NaI(Tl) crystal. The photomultiplier tube has 10 dynodes, with each dynode providing an electron multiplication factor of 3. About 80% of the light from the crystal is absorbed by the photocathode, which has a photocathode efficiency (number of electrons emitted per light photon absorbed) of 0.05. Assuming that 30 photons of light are produced in the NaI(Tl) crystal per kiloelectron volt of energy absorbed, compute the number of electrons received at the anode of the photomultiplier tube.
- *8-5. An α -particle from ^{210}Po (5.30 MeV) is absorbed completely in an air-filled pulse-type ionization chamber. Assuming a capacitance of 40 picofarads, compute the size of the voltage pulse.
- 8-6. Compare the efficiency and resolution of a semiconductor detector and a scintillation detector exposed to γ rays. Why are the efficiency and resolution different?

*For problems marked with an asterisk, answers are provided on p. 491.

SUMMARY

- An ionization chamber consists of parallel plate or coaxial electrodes in a volume occupied by a filling gas. The gas is sometimes air at ambient temperature and pressure.
- Quantities measured with an ionization chamber include:
 - Size of voltage pulse
 - Total charge collected
 - Magnitude of electrical current
- Uses of ionization chambers in radiology and nuclear medicine include:
 - Isotope calibrators
 - Gaseous sample counting
 - Portable survey meters
 - Neutron detection (with BaF_3 gas or lithium, or boron electrode coating)
- Proportional counters employ the principle of gas amplification to increase the magnitude of the signal.
- Geiger counters use further increases in voltage to obtain complete discharge of the gas in the counting volume with each counting event. Because all detection events elicit the same response, energy resolution is sacrificed to achieve a greater pulse size.
- Solid scintillation detectors produce light in response to absorption of ionizing radiation. Scintillation crystals are often coupled with photomultiplier tubes to achieve a measurable electrical signal, but may also be coupled with diode/thin film transistor arrays.
- Liquid scintillation employs the same principles of solid scintillation, but can be used to study dissolved samples of low-energy beta emitters.
- Semiconductor detectors record ionization events within their depletion regions as bulk current or as a voltage pulse.

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